

Price Levels, Nonlinear Contracts, and Medical Decisions^{*}

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Abstract

We show that nonlinear pricing terms in vertical contracts significantly affect medical decisions. We compare fee-for-service (FFS) versus bundled payment (BP) contracts between insurers and healthcare practices focusing on medical decisions for childbirth in Colombia. We find that covering c-sections under FFS increases c-section rates by 34%; covering vaginal deliveries under FFS reduces c-section rates by 22%. Treatment effects are 50% larger when the switch induces larger expected price increases under FFS and when the zero marginal price under BP binds more. Despite shifts in procedure choices, maternal and infant outcomes are the same under both contracts.

Keywords: Fee-for-service, Bundled payments, Moral hazard, Cesarean section.

JEL codes: I11, I13, I18.

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1 Introduction

Relationships between upstream and downstream firms are often governed by complex contracts specifying nonlinear pricing schedules (Lee et al., 2021; Katz, 1989). When these nonlinear features do not bind, vertical relationships can be summarized by linear prices, i.e., a constant payment per unit of output. While this linear approximation is valid in several contexts (e.g., Ho and Lee, 2017; Gowrisankaran et al., 2015; Draganska et al., 2010), recent evidence shows that nonlinear pricing can significantly shape market outcomes (Ho and Lee, 2024; Bonnet et al., 2013; Conlon and Mortimer, 2013; Crawford and Yurukoglu, 2012; Rennhoff and Serfes, 2009; Mortimer, 2008). Separating the effects of changes in price levels from changes in the price schedule is empirically difficult, yet essential for understanding whether linear prices adequately capture incentives.

In this paper, we study how nonlinear contracts between health insurers and healthcare practices (hospitals and clinics) impact medical treatment decisions. We compare the impact of two contracts – fee-for-service (FFS) and bundled payments (BP) – on treatment decisions for childbirth. This setting is well-suited for our analysis because there are two distinct delivery methods – cesarean section (c-section) and vaginal delivery – allowing us to characterize medical practice for childbirth.

Our empirical setting is Colombia’s health system, where multiple insurers negotiate with multiple healthcare practices (hospitals and clinics) over contracts for every health service. Insurers and practices may cover c-sections and vaginal deliveries under different contracts, and may even index these contracts to patient characteristics, creating contract variation within insurer, practice, and type of woman. For example, insurer j and practice h can cover c-sections for women under the age of 27 under BP, but can cover the same procedure for women aged 27 or older under FFS. We denote the combination of insurer, practice, and type of woman as the “contracting unit.”

Under these conditions, each contracting unit has four possible contract combinations for c-sections and vaginal deliveries: FFS-FFS, FFS-BP, BP-FFS, and BP-BP. Most notably, we observe these contracts in our data, allowing us to exploit the highly granular variation within contracting unit to estimate the causal impact of contracts on c-section rates and to

separately identify the roles of price levels and nonlinear pricing incentives.

FFS and BP contracts differ mainly in whether prices are nonlinear. Under a FFS contract, the insurer and the practice negotiate a (linear) service price that is paid every time the service is provided. Under BP, they negotiate a (nonlinear) price for an episode of care involving a fixed number of services. If the practice provides more services during the episode than what the contract specifies, it does not receive payment for those additional services. For example, a BP contract may cover one c-section and one hospitalization day. If a woman covered under this contract stays two days in the hospital, the insurer does not reimburse the practice for the second day. Hence, prices for marginal ancillary services under BP are zero. Moreover, delivery procedure prices may differ between FFS and BP. A switch from BP to FFS therefore changes practices' incentives along two distinct margins: the level of the delivery price and the marginal price of ancillary services, allowing us to separate the contribution of price levels from that of nonlinear pricing.

Variation across procedures in both the level of prices and their nonlinear structure can shape downstream medical decisions. Under BP, the practice bears the risk of providing services that will not be reimbursed, and may therefore discourage the delivery procedure covered under BP, depending on how the alternative procedure is reimbursed. We develop a stylized theoretical model that predicts how c-section rates change following transitions between contracts, and we test these predictions in the data.

We use individual-level data covering nearly half of all deliveries in Colombia between 2013 and 2019. We aggregate the data to the level of a contracting unit and implement an event study approach, comparing trends in c-section rates between contracting units that switch their contract for a single delivery procedure at different moments in time. Identification of the causal effect relies on the assumption that the timing of contract switches is unrelated to pre-switch trends in c-section rates and that there are no other changes that occur contemporaneously with contract switches. We provide evidence supporting these assumptions, including tests of whether contracting units exhibit parallel trends prior to the switch. Importantly, the common concern of selection of women into contracts is minimized in our setting since we exploit contract variation within type of

woman.¹

The possibility that contracts differ between c-sections and vaginal deliveries within the same contracting unit presents both empirical challenges and opportunities. On the one hand, correctly identifying the treatment effect of a c-section contract switch requires holding the vaginal delivery contract fixed to avoid confounding effects. On the other hand, this variation allows us to estimate treatment effects for c-section and vaginal delivery contracts separately, providing a complete characterization of medical decisions for childbirth.

Our main findings show that contracts have meaningful effects on delivery procedure choice. We find that c-section rates increase by 17 percentage points (p.p.) (34%) when c-sections switch from BP to FFS (with vaginal deliveries fixed at BP) and decrease by 11 p.p (22%) when vaginal deliveries switch from BP to FFS (with c-sections fixed at BP).

To assess the relative importance of price levels and the nonlinear price schedule, we conduct two analyses. First, we estimate treatment effects as a function of the predicted change in delivery price induced by the switch to FFS, and find that c-section rates increase substantially more among insurer-practice pairs facing larger predicted price increases. Second, we estimate treatment effects as a function of spending on ancillary services, the margin on which BP's nonlinearity binds. We find that treatment effects are markedly larger for contracting units with higher baseline spending on ancillary services. Together, these results indicate that substitution away from procedures reimbursed under BP is driven by both price levels and nonlinear pricing incentives.

We next characterize the marginal woman affected by these incentives by estimating treatment effects as a function of high-risk pregnancy rates, fraction of women with above-average age, chronic disease prevalence, and rates of placenta previa/abruptia, the latter of which is a clinical indicator of c-section suitability ([Corredor-Waldron et al., 2024](#)). Across all measures of maternal health, treatment effects are approximately 50% larger among insurer-practice pairs serving sicker women relative to pairs that serve healthier ones. In particular, heterogeneity by rates of placenta previa/abruptia supports the interpretation

¹Other work has studied hospital selection into government contracts in the context of hip and knee replacements ([Einav et al., 2022](#)).

that, following a switch from BP to FFS, the marginal woman receiving a c-section is more clinically suited for the procedure. These results suggest that contract design can better align financial incentives with patient need.

We also examine heterogeneity by practice size and find that responsiveness to financial incentives is more pronounced among larger practices, where scale economies may facilitate investment in administrative capacity to monitor and respond to financial incentives. Overall, the findings are consistent with practice-level moral hazard: contracts with insurers influence medical decisions in ways that are not contractible or fully observable at the time of contracting. A natural question is whether these contract-induced changes in medical decisions affect health outcomes. We find no evidence that FFS increases rates of severe maternal morbidity or worsens infants' 5-minute APGAR scores relative to BP; if any, reductions in c-section rates are accompanied by lower maternal morbidity.²

Our paper makes four contributions to the literature. First, we provide causal estimates of the effect of FFS and BP on treatment decisions. Existing evidence has mainly documented associations between FFS use and healthcare spending (e.g., [Baker, 1999, 1997](#)), analyzed system-wide reforms such as transitions from FFS to managed care (e.g., [Kuziemko et al., 2018](#); [Aizer et al., 2007](#)), or leveraged variation in FFS prices or in the degree of capitation separately (e.g., [Gruber et al., 1998](#); [Grant, 2009](#); [Clemens and Gottlieb, 2014](#); [Ho and Pakes, 2014](#)). In contrast, we exploit granular variation in contracts within insurer-practice-woman-type cells. Second, because both the delivery price and the marginal price of ancillary services change when contracts switch, we can separate the effects of price levels from those of nonlinear pricing – margins that prior designs cannot jointly identify. Third, we study a novel setting in which incentives operate at the practice level rather than the individual physician level, complementing experimental evidence on payment contracts for maternity care providers in India ([Mohan et al., 2021](#)) and structural work on optimal dialysis medication contracts in the US ([Gaynor et al., 2023](#)). Finally, by showing that payment contracts matter for delivery procedure choice, we contribute to the literature on the determinants of c-section rates, which has examined the

²The APGAR score evaluates five criteria (Appearance, Pulse, Grimace, Activity, Respiration) at one and five minutes after birth, with higher scores indicating better infant condition.

impacts of place (e.g., [Fischer et al., 2024](#); [Robinson et al., 2024](#); [Card et al., 2023](#)), diagnostic expertise (e.g., [Corredor-Waldron et al., 2024](#); [Currie and MacLeod, 2017](#)), and physician moral hazard (e.g., [Foo et al., 2017](#); [Johnson and Rehavi, 2016](#); [Allin et al., 2015](#); [Godager and Wiesen, 2013](#)).

The remainder of this paper is structured as follows: Section 2 describes our institutional setting and presents the theoretical framework, Section 3 describes our data, Section 4 presents our empirical design, Section 5 shows our main results, and Section 6 concludes.

2 Setting and Theoretical Framework

We study the contracts established between insurers and healthcare practices in Colombia’s contributory healthcare system, which covers the half of the population in the country who pay payroll taxes and their dependents (around 25 million people). In this system, private insurers provide one national health insurance plan to their beneficiaries through a network of preferred providers. Other elements of the insurance plan such as premiums, cost-sharing, and benefits are strictly regulated and uniform across insurers and practices. Healthcare practices in our setting include hospitals and clinics, but not individual doctors.

Insurers negotiate with practices over contracts for each health service covered by the national insurance plan. This means that different services such as c-sections and vaginal deliveries may be covered under different contracts even for the same insurer-practice pair. Insurers and healthcare practices can choose from a set of three contracts: FFS, BP, and capitation. If a service is covered under FFS, the insurer negotiates a price to be paid every time the service is provided to one of its beneficiaries only after it has been provided. Thus, FFS is fully retrospective and the price of a marginal service is strictly positive.

If a service is covered under capitation, the insurer negotiates a price per person, a service frequency per person, and a target population, and pays the healthcare practice the product of these three elements before services are provided to their beneficiaries, typically at the beginning of the calendar year. Thus, capitation is fully prospective and the price of a marginal service is zero. Low-complexity services such as primary care

consultations and blood tests tend to be capitated.³

Under BP, the insurer negotiates a price for a bundle of services used during an episode of care. This price is paid every time the episode of care occurs. Episodes of care typically covered under BP tend to be highly standardized such as hysterectomies, appendectomies, or deliveries. BP is less than fully retrospective because the insurer does not pay for additional services provided during the episode of care that are not included in the bundle. For example, if the BP contract covers one hospitalization day for a c-section but the woman stays two days at the hospital, this additional day is not reimbursed by the insurer. Hence, the price for marginal ancillary services included in the bundle (along with the delivery procedure) is zero.

FFS contracts differ from BP and capitation in both price levels and whether prices for marginal services are strictly positive or zero. Therefore, for the remainder of this study we will distinguish between FFS and non-FFS, where non-FFS refers to both capitation and bundled payments and will be referred to as simply BP.

Throughout our analysis, we assume physician and practice incentives are aligned. Colombian physicians are often employed at a single institution, allowing the practice to more directly influence their behavior.⁴ Under this assumption, physicians at healthcare practices with FFS contracts have an incentive to provide the delivery procedure with the highest profit margin, while those at practices with BP contracts have an incentive to provide the lowest-cost procedure since payments do not vary with the number of services actually rendered. These incentives have been well documented in other settings where contracts vary over time but do not vary across services (e.g., [Kuziemko et al., 2018](#); [Adida et al., 2017](#); [Ho and Pakes, 2014](#)). However, in our setting where contracts may differ across services, healthcare practice incentives will depend on the combination of contracts across delivery procedures.

To see these incentives more clearly, we propose a simple model in which the practice chooses a procedure for each woman to maximize profits. This decision is independent

³Low complexity services, such as vaginal deliveries, are more likely to be covered under capitation.

⁴Hospitalists and OB/Gyns at private healthcare practices or hospitals are typically paid a base salary plus commissions per procedure or are paid directly per procedure. In the case of public hospitals, physicians and OB/Gyns tend to have fixed salaries.

across women and applies to all physicians within the practice.⁵ Suppose each woman has health status h . Higher values of h denote that the woman is in worse health. For c-section, let the FFS price be p_C , the BP price be b_C , and the marginal cost be $m_C(h)$. The corresponding values for vaginal deliveries are p_V , b_V , and $m_V(h)$. The BP price reimburses the practice not only for the provision of the delivery procedure, but also for ancillary services included in the bundle, the marginal cost of which is $m_O(h)$. For simplicity, we assume the BP contract for c-sections and vaginal deliveries includes the same set of ancillary services. In this setting, FFS and BP contracts differ in price levels (e.g., p_C versus b_C) as well as in prices for marginal ancillary services, which make BP contracts nonlinear. Finally, we assume all marginal cost curves are monotonically increasing in health status.⁶

We assume the marginal cost of a c-section increases more rapidly with the woman's health status than that of a vaginal delivery ($0 < m'_V(h) < m'_C(h)$) to reflect the fact that surgical procedures are more complex among patients in worse health. This assumption holds in our data as seen in Appendix Table 1, where we show that the difference in healthcare spending on the delivery day between women with and without chronic conditions is higher for those receiving c-sections than those receiving vaginal deliveries. We also assume practices are risk neutral, although our results would extend to settings with risk aversion if we assume marginal cost curves are strictly convex.

The profit from choosing procedure $s \in \{C, V\}$ conditional on the contracts and the woman is

$$\Pi_s = E[\pi_s | \mathbf{FFS}, h] D(h)$$

with

$$E[\pi_s | \mathbf{FFS}, h] = (p_s - m_s(h)) \text{FFS}_s + (b_s - (m_s(h) + m_O(h)d_O))(1 - \text{FFS}_s)$$

⁵In this model, we also take insurer-practice contracting decisions as fixed and focus on practices' downstream procedure choices. In the upstream contracting problem, a risk-neutral insurer contracts with a practice whose treatment intensity is unobservable to the insurer, generating a problem of asymmetric information. The treatment intensity determines the probability of c-sections and vaginal deliveries. The insurer offers a contract to the practice, which it can accept or reject. Our analysis centers on determining the practice's optimal choice of treatment intensity, given that they have accepted the contract.

⁶Our specification of the BP contract closely follows [Einav et al. \(2022\)](#).

where FFS_s is an indicator for whether service s is covered under FFS and d_O is the number of ancillary services provided. Here we have assumed practice demand $D(h)$ is independent of the contracts (ruling out risk selection) since these are not directly observable to women before choosing where to give birth.⁷ The practice's problem is

$$\max_{s \in \{C, V\}} \{E[\pi_C | \text{FFS}, h], E[\pi_V | \text{FFS}, h]\}$$

The trade-off for a practice in choosing a procedure covered under BP is that it may receive a higher payment ($b_s > p_s$) but receives no additional reimbursement for marginal ancillary services supplied as part of the episode of care.

TABLE 1: Contracting Scenarios and Incentives

Scenario (1)	Contract (2)	Service (3)	Expected practice profit (4)
1	BP	C-section	$b_C - (m_C(h) + m_O(h)d_O)$
	BP	Vaginal delivery	$b_V - (m_V(h) + m_O(h)d_O)$
2	FFS	C-section	$p_C - m_C(h)$
	FFS	Vaginal delivery	$p_V - m_V(h)$
3	FFS	C-section	$p_C - m_C(h)$
	BP	Vaginal delivery	$b_V - (m_V(h) + m_O(h)d_O)$
4	BP	C-section	$b_C - (m_C(h) + m_O(h)d_O)$
	FFS	Vaginal delivery	$p_V - m_V(h)$

Note: Table shows the practice's profit under four possible contracting scenarios in which c-sections and vaginal deliveries are either covered under the same contract or under different contracts. The FFS price, bundled payment, and marginal cost of a c-section are p_C , b_C and $m_C(h)$, respectively. The corresponding values for vaginal delivery are p_V , b_V and $m_V(h)$. d_O is the number of ancillary services provided during the delivery. h denotes the woman's health status, with higher values implying worse health.

Table 1 summarizes practices' expected profit per woman under the four possible contract combinations ("contracting scenarios" hereafter). The table highlights that the optimal procedure choice in each contracting scenario will follow a cutoff rule in h . For example, in scenario 2 where both procedures are covered under FFS, the health status h^* at which the practice is indifferent between providing a c-section and a vaginal delivery is implicitly defined by $p_C - m_C(h^*) = p_V - m_V(h^*)$. This equation gives a unique solution

⁷Patients in Colombia do not observe the contracts between insurers and practices, but may select into practices based on health status, which may in turn be correlated with use of FFS. Thus, our assumption is that conditional on h , practice demand does not vary with the contracts.

for h^* given our assumption of monotonicity of the marginal cost curves.

The table also provides insight into how to compute the effect of each contract. For example, to estimate the effect of covering c-section under FFS we can compare scenario 3 to 1, where the contract for vaginal deliveries is fixed to BP. Moreover, to estimate the effect of covering vaginal delivery under FFS we can compare scenario 4 to 1, where the contract for c-sections is fixed to BP.

The impact of the different contracting scenarios on c-section rates is generally ambiguous and depends on the relative magnitudes of the difference in prices levels ($b_s - p_s$) and the cost of ancillary services ($m_O(h)d_O$). For example, in the case of scenarios 1 and 3, switching c-section from BP to FFS results in a decline in prices of $b_C - p_C$, but also a decline in costs of $m_O(h)d_O$. C-section rates will only rise after switching to FFS if the potential decline in delivery prices is offset by the potential cost savings from unbundling ancillary services.

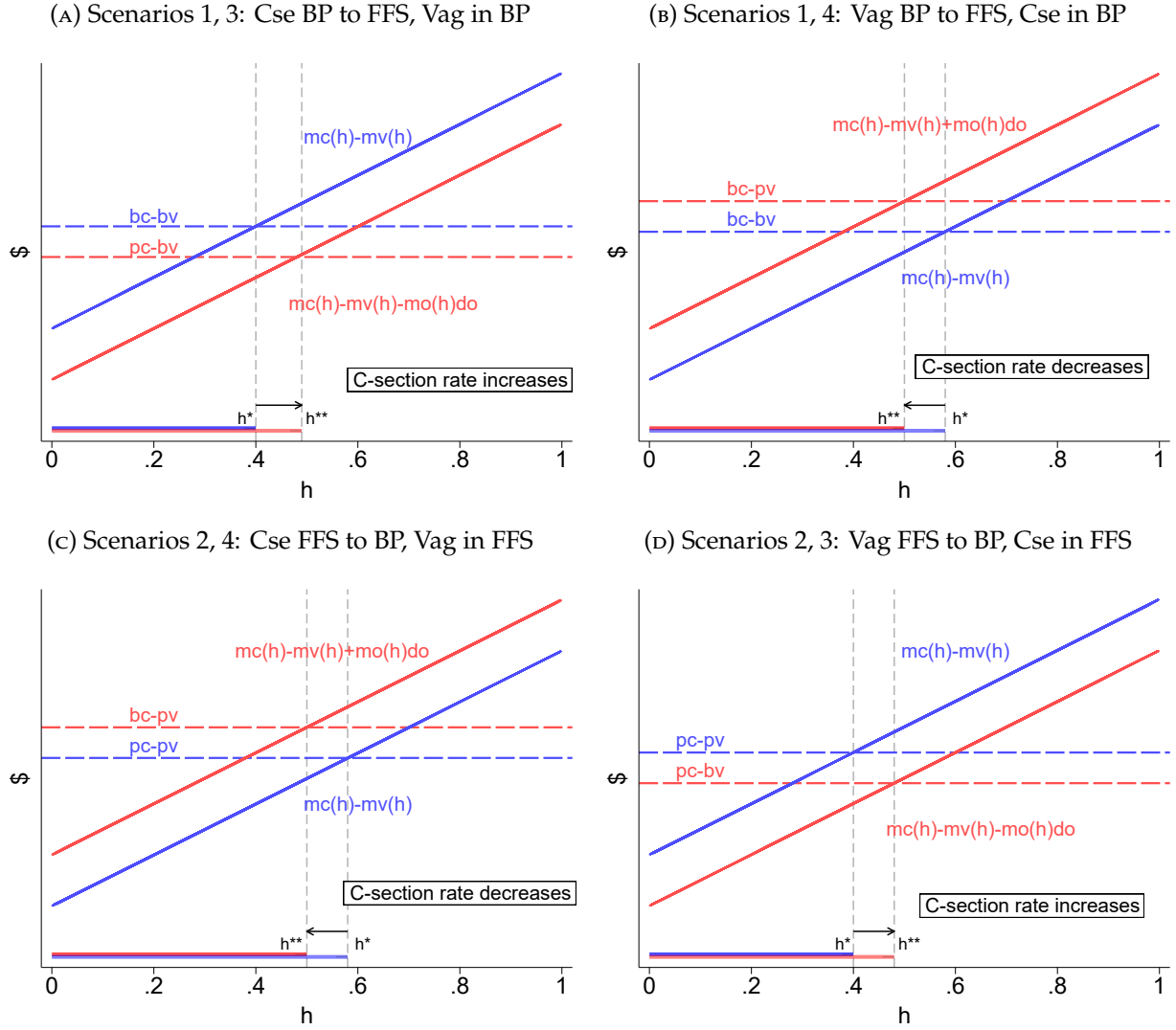
Figure 1 illustrates this case. Panel A depicts the comparison of scenarios 1 and 3. The blue lines correspond to scenario 1 where both procedures are covered under BP and the red lines correspond to scenario 3 where c-sections are covered under FFS and vaginal deliveries under BP. The solid blue line shows the difference in total cost between c-section and vaginal delivery in scenario 1. The common cost of ancillary services $m_O(h)$ cancels out of this difference, which is increasing in health status given our assumption on $m'_s(h)$. The dashed blue line represents the difference in prices between the two procedures. Under these conditions, h^* is the health status below which the practice chooses to provide a c-section.

The solid red line in Panel A depicts the difference in cost between c-section and vaginal delivery in scenario 3. This curve is shifted to the left relative to the solid blue line because under BP the practice bears the cost of ancillary services. The dashed red line represents the difference in prices between the two procedures, and h^{**} is the cutoff health status for this combination of contracts.

As long as the difference between b_C and p_C is smaller than the difference between costs, our prediction is that $h^{**} > h^*$. Thus, the range of health statuses below which the practice chooses a c-section increases when going from scenario 1 to 3. This means that a

woman is more likely to receive a c-section when the contract for c-section switches from BP to FFS (holding vaginal delivery fixed at BP), and that the marginal woman receiving a c-section is relatively sicker. Finally, note that the higher the FFS price of a c-section, the higher the increase in c-section rates.

FIGURE 1: Effects of Transitioning between Contracting Scenarios



Note: Figure presents a graphical representation of our model predictions from Table 1. Panel A compares the health status cutoffs at which the practice is indifferent between providing a c-section and a vaginal delivery for contracting units in scenario 1 where both procedures are covered under BP and scenario 3 where c-section is covered under FFS and vaginal delivery under BP. Panel B compares the cutoffs in scenario 1 where both procedures are covered under BP and scenario 4 where c-section is covered under BP and vaginal delivery under FFS. Panel C compares the cutoffs in scenario 2 where both procedures are covered under FFS and scenario 4 where c-section is covered under BP and vaginal delivery under FFS. Panel D compares the cutoffs in scenario 2 where both procedures are covered under FFS and scenario 3 where c-section is covered under FFS and vaginal delivery under BP. h^* denotes the health status below which women receive c-sections under the original contracting scenario, and h^{**} denotes the health status below which women receive c-sections under the new contracting scenario.

Panel B depicts the cutoff values for scenarios 1 and 4, indicating that the likelihood of receiving a c-section decreases when the vaginal delivery contract switches from BP to FFS (while c-sections are fixed at BP). In this case, the marginal woman receiving a c-section is in relatively better health. Panel C shows the comparison between scenarios 2 and 4, predicting that the c-section rate should decrease when the c-section contract switches from FFS to BP (holding vaginal delivery fixed at FFS). Finally, Panel D predicts that the c-section rate should increase when moving from scenario 2 to 3 where the contract for vaginal delivery switches from FFS to BP (with c-section fixed at FFS).

In our empirical application, we take advantage of variation in contracts over time and across procedures to estimate whether practices' responses are consistent with the predictions of our model. Failure to account for differential responses to FFS by delivery procedure can lead us to underestimate the desired treatment effect. For example, comparing switches from scenario 4 to 3 would likely result in an increase of the c-section rate since c-sections are covered under FFS in scenario 3. However, this naïve comparison confounds the impact of changes in the contract for vaginal deliveries. For example, if vaginal deliveries are more likely to be provided in scenario 4, then the effect of the switch in c-section contracts on c-section rates will be biased towards zero.

We note that our model is a simplified representation of real-world medical decisions and may not capture several other ways contracts may influence medical decisions. For example, physicians at these healthcare practices may be altruistic, considering both profits and patient utility when making procedure choices. Although we do not consider patient utility in the medical decision, if the degree of altruism does not vary across contracting scenarios, then altruism will not affect the predictions of our model.

As another example, practices under BP may have incentives to select low-risk patients who are less likely to need health care or insurers may have incentives to steer high-risk patients towards practices covered under BP. These risk selection incentives can make healthcare practice demand vary with the contracts, which would lead us to underestimate the impact on c-section rates. For example, if insurers send all the high-risk women who potentially need c-sections to practices where c-sections are covered under BP, then the impact of FFS on the c-section rate will be biased toward zero. Our empirical design

addresses this common selection concern by leveraging within-woman type variation in contracts. We will also directly explore whether contracts generate risk selection incentives in Section 5.5.

There may be other unobserved determinants of delivery procedure choice such as convenience or practice-level capacity constraints, which we do not consider. However, if moral hazard at the practice level is the predominant way in which contracts affect delivery procedure choice and outcomes, we can expect to estimate results in line with the predictions of our model.

3 Data and Descriptives

We use health claims data from all pregnant women in the contributory system who had a childbirth between 2013 and 2019, covering nearly half of all births in the country. These data report the type of service, service date, healthcare practice, insurer, negotiated service price, and contract for each claim. There are 1,017,830 deliveries in the raw data. We categorize women as having a high-risk pregnancy if they received a diagnosis indicating that their pregnancy was high risk any time during the 9 months of pregnancy.⁸ We also categorize women as having severe maternal morbidity (SMM) based on the services and diagnosis codes rendered between 1 week and 1 month after childbirth following the Centers for Disease Control and Prevention (CDC)'s definition.⁹ A woman is defined as having a chronic disease if she received any of the following diagnoses before her delivery: diabetes, cancer, cardiovascular disease, renal disease, pulmonary disease, genetic diseases, epilepsy, HIV-AIDS, transplants, and tuberculosis. We follow the ICD categorization in [Riascos et al. \(2014\)](#) to construct these diagnosis groups.

We focus on reproductive-age women (15-50), exclude women with multiple or breech pregnancies for whom c-sections are medically recommended, and keep nulliparous (or

⁸These ICD-10 codes include: supervision of high-risk pregnancies (O09), infections of genitourinary tract in pregnancy (O23), pregnancy-induced hypertension or pre-eclampsia (O10-O16), hemorrhage due to threatened abortion (O20), excessive vomiting during pregnancy (O21), venous complications and hemorrhoids in pregnancy (O22), diabetes mellitus in pregnancy, childbirth, and the puerperium (O24), malnutrition in pregnancy, childbirth and the puerperium (O25), abnormal findings on antenatal screening of mother (O28), and complications of anesthesia during pregnancy (O29)

⁹See [Centers for Disease Control and Prevention \(2024\)](#).

first-time) deliveries.¹⁰ After dropping deliveries with missing contract data or multiple contract switches, which we discuss further below, our final analysis sample contains 778,510 deliveries. See Appendix A for a summary of these data cleaning steps. Moreover, we do not observe the specific physician that performed each service, nor do we observe whether a c-section is scheduled or emergent.

TABLE 2: Summary Statistics of Analysis Sample

	C-sections		Vaginal deliveries	
	Mean	SD	Mean	SD
Fee-for-service (FFS)	0.665	(0.472)	0.608	(0.488)
Bundled payment	0.296	(0.457)	0.363	(0.481)
Pure capitation	0.039	(0.193)	0.028	(0.166)
Age≤27	0.473	(0.499)	0.590	(0.492)
Age>27	0.527	(0.499)	0.410	(0.492)
High pregnancy risk	0.434	(0.496)	0.371	(0.483)
Severe maternal morbidity (SMM)	0.050	(0.217)	0.030	(0.171)
Unhealthy	0.391	(0.488)	0.339	(0.473)
Gestational age (weeks)	38.01	(2.041)	38.58	(1.537)
5-min APGAR<8	0.018	(0.131)	0.010	(0.099)
Spending on delivery day [†]	535.5	(536.3)	419.2	(277.1)
Spending up to delivery day [†]	467.9	(632.7)	389.4	(461.5)
Deliveries	357380		421130	
Insurer-provider pairs	1954		1949	
Providers	689		687	
Insurers	15		15	

Note: Table shows the mean and standard deviation (in parenthesis) of the main variables in our analysis sample separately by delivery procedure. An observation is a delivery. The 5-minute APGAR score reflects how well the infant tolerated the birthing process and is measured on a scale from 1 to 10, with a higher score reflecting better infant health. ([†]) Measured in USD of 2014.

Table 2 presents summary statistics of our analysis sample. An observation is a delivery claim. The c-section rate equals 46%. Most deliveries are reimbursed under a FFS contract, and FFS use is more prevalent for c-sections than for vaginal deliveries. Among less retrospective contracts, we see that most insurer-practice pairs tend to use bundled payments over capitation, and that the use of these contracts is more common for vaginal deliveries. More than half of vaginal deliveries are performed on women aged 27 (the average age in our sample) or younger, while more than half of c-sections are among

¹⁰We identify breech pregnancies using ICD-10 code O32 and multiple pregnancies using procedure codes from the Colombian national insurance plan: 735930 (assistance with spontaneous twin or multiple birth) and 735931 (assistance with an intervened twin or multiple birth). To identify nulliparous deliveries we search over our 7 years of data the woman's first delivery procedure.

women over 27 years old. Less than half of pregnancies are classified as high risk, and the rate of high-risk pregnancies is higher among c-sections than vaginal deliveries. Likewise, spending up to and on the day of the delivery is higher for c-sections. Health outcomes such as the infant’s likelihood of having a 5-minute APGAR score below 8 and the mother’s gestational age are marginally better for vaginal deliveries.

3.1 Obtaining Contracts from the Data

Insurers and healthcare practices typically negotiate contracts at the beginning or the middle of every calendar year, but when these negotiations end and new contracts go into effect can vary. Insurer-practice pairs negotiate contracts for each delivery procedure and may choose to index contract types to patient characteristics. Appendix Figure 1 shows screenshots of contracts between insurers and public hospitals obtained from publicly available sources. Contracts for delivery procedures tend to be indexed to the patient’s level of “complexity,” hence we characterize delivery procedure contracts along variables that describe the woman’s level of complexity such as her age and pregnancy risk.

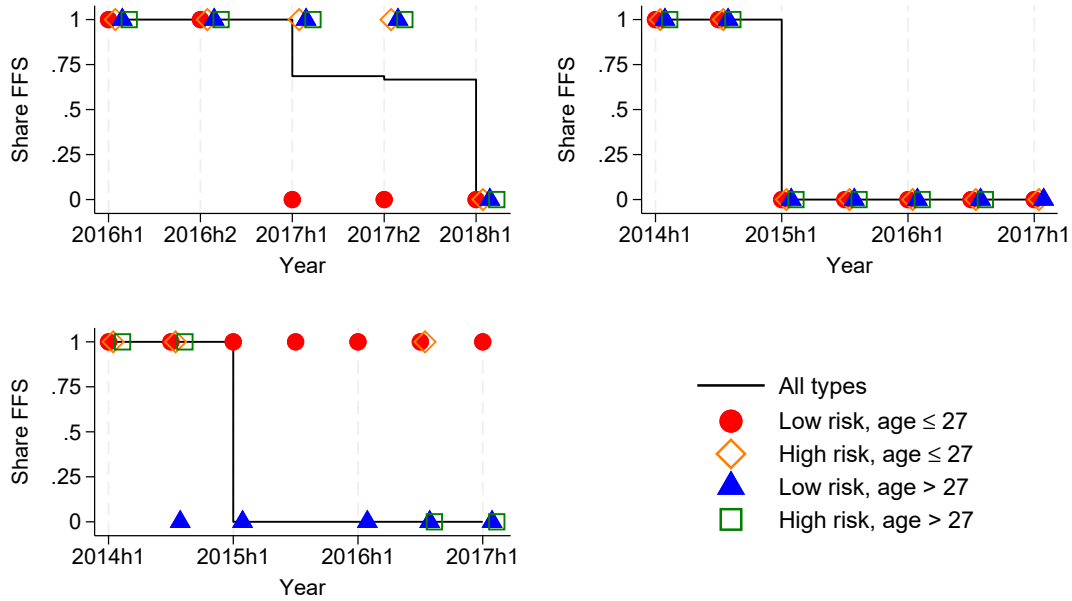
In the claims data, we observe only the contract for the procedure actually provided. For example, if woman i has a vaginal delivery, we observe her vaginal delivery contract but not the c-section contract she would have faced. We therefore recover the counterfactual contract from the claims of comparable women – those with the same age and pregnancy risk as i , treated by the same insurer-practice pair – who received the alternative procedure.

Specifically, we define a woman type as a combination of age group (≤ 27 , > 27) and pregnancy risk (high vs. low, as defined above); the age cutoff corresponds to the average age in the analysis sample, and our results are robust to alternative cutoffs. We then assign to each contracting unit in each half-year its modal contract for each delivery procedure. This procedure requires that, within a contracting unit and half-year, at least one woman received each of the two delivery procedures. This condition holds for around 50% of contracting units, which account for 97% of deliveries over the sample period; the remaining units are ones in which all women of a given type received the same procedure

in a given half-year, so the contract for the alternative procedure is never observed in claims. Reassuringly, our results are robust to dropping contracting units whose modal contract differs from the observed contract.

3.2 Trends in Contracting Decisions

FIGURE 2: Examples of C-section Contract Variation within Insurer-Practice Pairs



Note: Figure shows the contract for each type of woman within an insurer-practice pair between 2013 and 2019. Each panel corresponds to a different pair. The solid black line shows the share of c-sections in each half year that are reimbursed under FFS. The markers show the contract type for each woman type in each half year.

The primary source of variation we use in our empirical analyses is the temporal variation in contracts within insurer-practice pair for a given woman type. Figure 2 illustrates this variation in the c-section contract for three of the 1,965 insurer-practice pairs in our analysis sample. The top right panel shows an insurer-practice pair where all c-sections were reimbursed under FFS before 2015-1 but switched to BP thereafter. Contracts also vary across woman types. For instance, the top left panel shows an insurer-practice pair where c-section was reimbursed under FFS for all woman-types until 2017, when low-risk women aged 27 or less switched to BP.

FIGURE 3: Trends in the Use of FFS for Delivery Procedures

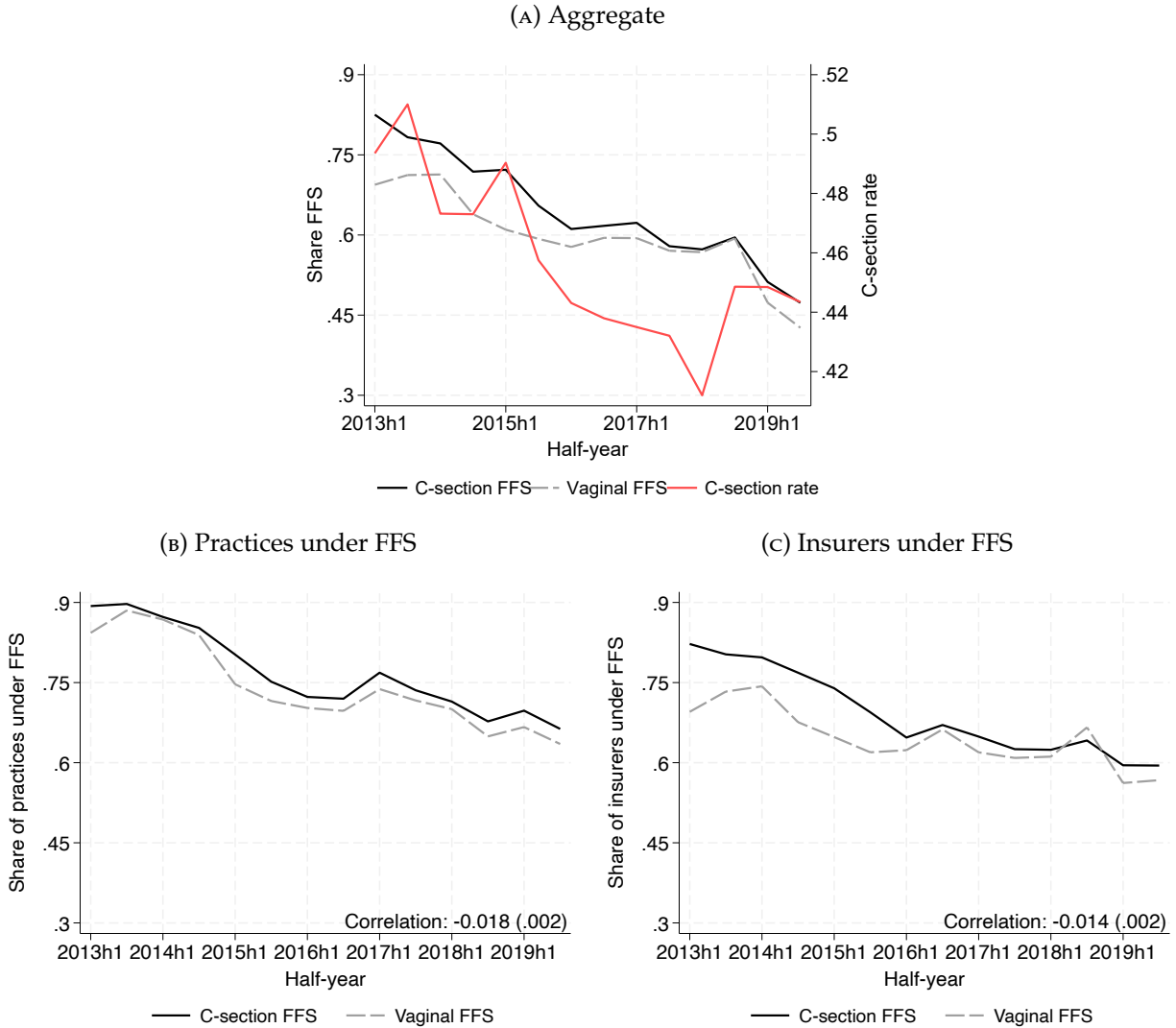


Figure 3 illustrates aggregate trends in the use of FFS. Panel A shows that, in aggregate, use of FFS fell over the sample period by approximately 30 p.p. from a baseline of 85% for c-sections and 70% for vaginal deliveries. At the same time, the c-section rate in the contributory system declined from 50% to 40% between 2013-1 and 2018-1 – a pattern confirmed by studies in the medical literature (e.g., [Zuleta-Tobón and Jairo, 2023](#)).

The contract trends are explained both by insurers switching most of the healthcare

practices in their network to BP as seen in Panel B, and by practices increasing the number of insurers that they contract with under BP as seen in Panel C. The decreasing trend in the use of FFS is slightly stronger among practices within an insurer than among insurers within a practice, as seen by the correlation coefficient in each panel. Given that BP contracts tend to transfer the financial risk to the practice, the stronger correlation among practices within an insurer suggests the insurer holds the bargaining power in the context of delivery procedures.

4 Empirical Design

When contracts are established at the procedure level, it is possible that c-sections are more likely to be covered under FFS than vaginal deliveries across all insurer-practice pairs. It is also possible that women for whom c-sections are medically recommended are more likely to be covered under FFS. This suggests that a comparison of c-section rates *across* insurer-practice pairs would yield a mechanically higher c-section rate under FFS than BP because of selection of women and practices into contracts. Thus, the appropriate research design should instead exploit the variation in contracts *within* procedure, insurer, and practice, and type of woman, which the Colombian setting uniquely provides.

We estimate event study models as follows:

$$y_{jhmt} = \sum_{\substack{k=-5 \\ k \neq -1}}^4 \alpha_k 1\{t - t^* = k\} \times \text{FFS}_{jhmt}^s + \gamma_{jm} + \delta_t + \varepsilon_{jhmt} \quad (1)$$

where y_{jhmt} is an outcome for insurer j , practice h , and type of woman m in half year t ; FFS_{jhmt}^s is an indicator that procedure $s \in \{C, V\}$ switched to FFS; t^* is the half year when the contracting unit switches their contract; γ_{jm} are insurer-woman type fixed effects; and δ_t are half year fixed effects. The coefficients of interest are α_k , which measure the causal effect of reimbursing procedure s under FFS rather than BP in every half year relative to the switch.

To estimate these models, we collapse the claim-level data to the $jhmt$ -level, computing

per capita outcome measures. In our regressions, we examine outcomes around the *first* contract switch, dropping observations corresponding to subsequent contract switches, which yields an unbalanced panel. We cluster standard errors at the level of a contracting unit and use [Borusyak et al. \(2024\)](#)'s estimator to avoid potential biases from staggered contract switches.

Ideally, we would include contracting unit fixed effects to absorb all time-invariant heterogeneity at the insurer-practice-woman type level, but this is infeasible given that many practices are not continuously observed over time. Instead, we include insurer-woman type fixed effects, which absorb differences across insurers in how they contract for different types of women. This approach does not require continuous observation of every cell and aligns with the descriptive patterns suggesting insurers have relatively higher bargaining power when contracting over delivery procedures. However, it may raise issues related to changes in sample composition around the time of the contract switch.

We provide several robustness checks to address these issues such as specifications that include insurer-practice, insurer-half year, and woman type-half year fixed effects, which can help control for potential changes in insurers' relative bargaining power or enrollee composition over time that may be correlated with our outcome variables. We also explore whether there are changes in practice composition after the contract switch using practice characteristics as outcomes.

To map our empirical approach to the theoretical framework outlined in Section 2, we partition the data to the four contracting scenarios in Table 1. Table 3 presents the transition matrix between different contracting scenarios. Values in parentheses correspond to the number of deliveries in each cell, while values outside of parentheses are the number of observations in our analysis sample (i.e., contracting units-half years). We exclude from the table transitions that are irrelevant for our analysis, such as those where the contract for both c-sections and vaginal deliveries change.

The impact of covering c-sections under FFS is identified from the 344 observations (14,222 deliveries) that switch from scenario 1 to 3, where the contract for vaginal deliveries is fixed at BP and the contract for c-sections changes from BP to FFS. Focusing on the 195

observations (12,293 deliveries) that switch from scenario 1 to 4 allows us to identify the impact of covering vaginal deliveries under FFS. In both cases, the control group are the 2,030 observations (125,010 deliveries) always under scenario 1.

TABLE 3: Transition Matrix (C-section contract–Vaginal delivery contract)

	Scenario 1 BP-BP	Scenario 2 FFS-FFS	Scenario 3 FFS-BP	Scenario 4 BP-FFS
Scenario 1: BP-BP	2030 (125010)	— —	344 (14222)	195 (12293)
Scenario 2: FFS-FFS	— —	20784 (338392)	11 (45)	38 (309)
Scenario 3: FFS-BP	5 (194)	1048 (28268)	78 (170)	— —
Scenario 4: BP-FFS	4 (216)	31 (252)	— —	82 (227)

Note: Table shows the number of observations (contracting unit-half year) and the number of deliveries in parenthesis for every transition between contracting scenarios. Transitions in which the contract for both c-sections and vaginal deliveries change are excluded. The first contract reported in the contracting scenario corresponds to the c-section contract, the second corresponds to the vaginal delivery contract.

Note that, alternatively, we could estimate the impact of covering c-sections under FFS using contracting units that switch from scenario 4 to 2 where vaginal deliveries are fixed at FFS. However, in this case, there are only 31 treated observations corresponding to 252 deliveries. Similarly, we could estimate the effect of covering vaginal deliveries under FFS using units that switch from scenario 3 to 2, yet in this case the control group (units always under scenario 3) only contains 78 observations corresponding to 170 deliveries. For contract switches that identify the effect of moving towards BP (such as switches from scenario 2 to 4 or from scenario 2 to 3), we also have insufficient observations to conduct the analysis. Hence, given that our stylized model predicts the transition to and from FFS to be symmetric, our main empirical specifications comparing units that switch from scenario 1 to 3 or from scenario 1 to 4 will provide a comprehensive characterization of the impact of contracts on procedure choices.

Table 4 reports summary statistics for treated and control units in the pre-switch period.¹¹ The treatment groups are contracting units that switch between scenarios 1 and 3 in column (1) and between scenarios 1 and 4 in column (2). Control units are those

¹¹The number of observations and the number of deliveries in Table 4 differs from Table 3 because we focus only on pre-period data in the former but use all the data for the latter.

always under scenario 1. High-risk pregnancy rates and the fraction of women aged 27 or older differ between treatment and control groups, while maternal and infant outcomes are balanced. These differences in levels do not threaten our empirical design, which leverages changes within insurer-practice-woman types and thus requires only parallel trends rather than balance in levels. Moreover, we show in Appendix Figure 3 that our results are robust to dropping the control group entirely, so that identification relies solely on the staggered timing of contract switches among treated units.

TABLE 4: Summary Statistics

	Scenarios 1→3 Treated (1)	Scenarios 1→4 Treated (2)	Scenario 1→1 Control (3)
C-section rate	0.517	0.461	0.472
Age>27	0.471	0.658	0.501
Has high pregnancy risk	0.500	0.579	0.433
Has severe maternal morbidity (SMM)	0.038	0.054	0.028
Has chronic disease	0.308	0.326	0.276
Gestational age (weeks)	38.07	37.99	38.24
5-min APGAR<8	0.018	0.024	0.016
Spending on delivery day [†]	479.5	456.1	489.0
Spending up to delivery day [†]	464.2	521.0	461.1
Deliveries	12700	12126	123782
Observations	140	76	1996

Note: Table shows variable means for treated units in the pre-period for those that switching between scenarios 1 and 3 in column (1) and between scenarios 1 and 4 in column (2). Column (3) presents variable means for contracting units in the control group that never switch their contracts and are under scenario 1. ([†]) Measured in USD of 2014.

4.1 Identification Assumptions

Identification of the causal effect of contracts on delivery procedure choice relies on several assumptions. First, we require that the outcomes of contracting units that change their contracts would have evolved parallel to those that never change, had the switch not occurred. While insurers and practices likely select into contracts, this is not problematic for identification as long as contract switches are unrelated to pre-switch outcome trends (i.e., if selection is only on outcome levels). We provide evidence of this parallel pre-trends assumption by testing the statistical significance of pre-switch event study coefficients.

In our main specification, identification comes primarily from comparing treated contracting units that switch contracts at different points in time. This variation can yield biased estimates, however, if the timing of switches is non-random – for instance, if insurers switch contracts for higher-risk women earlier than for lower-risk women. Such staggered adoption driven by risk would manifest as dynamic treatment effects that trend over the post-period. Flat post-period dynamics would therefore be suggestive of timing of contract switches being orthogonal to underlying woman risk. We also provide more direct evidence that contract switches are unrelated to pre-switch characteristics below.

Second, we require that no other policy changes happen simultaneously with contract switches. The staggered nature of our contract switches reduces the risk that these switches coincide with other shocks. Still, it is possible that women’s unobserved preferences for c-sections change over time, prompting insurers to adopt BP contracts with practices in their networks. This could lead us to mistakenly attribute changes in c-section rates to the contract rather than to women’s unobserved preferences.¹² Also, if Colombian obstetricians become more skilled at performing c-sections in practices with FFS contracts, the observed effects could reflect unobserved improvements in provider ability rather than the contracts themselves. To address this, we estimate versions of equation (1) that include woman type-half year and insurer-half year fixed effects.

Third, we require that there are no spillover effects to the control group. We believe this is likely the case given that our control group is fairly large and includes insurer-practice pairs that operate in different geographical markets than those in the treatment group. Because contracts are the outcome of bilateral negotiations between insurers and practices, they are affected in equilibrium by the outcomes of all other insurer-practice pairs in the market. The fact that treated and control units may operate in different geographical markets minimizes the impact of these bargaining externalities.

In Table 5 we provide suggestive evidence of the quasi-random timing of contract switches. We regress an indicator for whether the contracting unit switches its contracts

¹²For example, Gaviria (2017) documents that differences in women’s attitudes toward c-sections account for part of the variation in c-section rates across Colombian states. The study reports that c-section rates among women with higher education are 15 p.p. higher than among those without formal education, controlling for health status and healthcare access.

on a wide range of lagged time-varying insurer and practice characteristics. As in our main specification, we include insurer-woman type fixed effects and half year fixed effects, and cluster standard errors at the contracting unit level. The results show that contract switches are generally unrelated to pre-switch outcomes (as seen by the joint F-test of significance), suggesting the direction of causality runs from contracts to outcomes, not the reverse. While we cannot definitively identify the reasons for switching contracts, plausible explanations include idiosyncratic managerial preferences, which studies have shown shape hospital behavior in other contexts ([Otero and Munoz, 2022](#))

TABLE 5: Correlates of Contract Switching

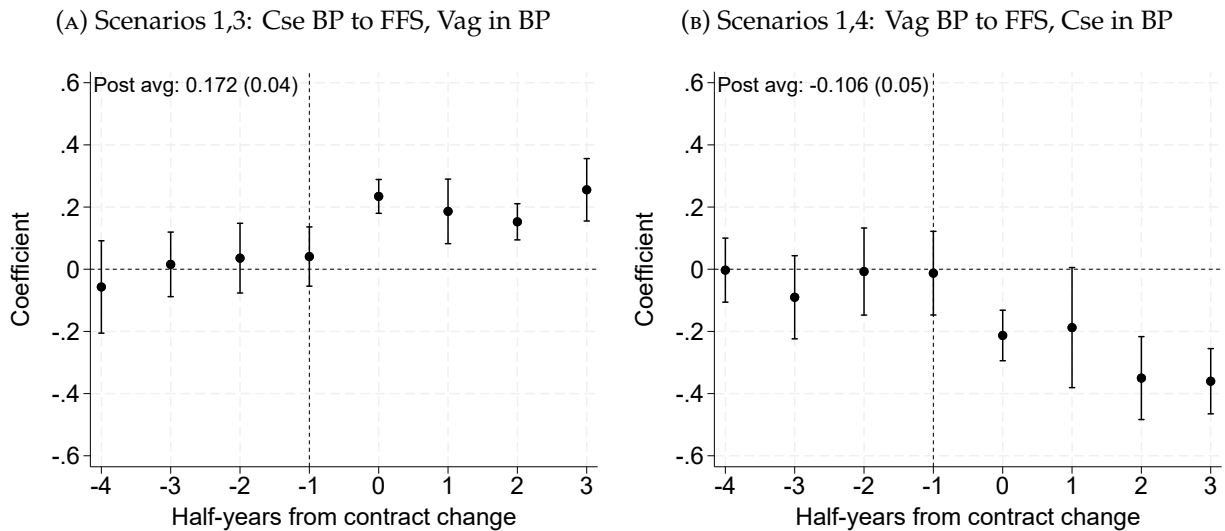
Lagged variables	Cse BP to FFS (1)	Vag BP to FFS (2)
C-section rate	-0.002 (0.015)	-0.014 (0.012)
SMM	-0.058 (0.043)	-0.012 (0.067)
Birth weight (kilograms)	0.011 (0.017)	-0.006 (0.014)
Chronic disease	-0.002 (0.019)	-0.011 (0.014)
1-min APGAR	-0.011* (0.006)	-0.002 (0.006)
Gestational age	-0.004 (0.004)	-0.006* (0.003)
Spending on delivery day	-0.005 (0.008)	-0.017* (0.009)
Spending up to delivery day	0.000 (0.004)	0.004 (0.005)
Woman-type share of deliveries	0.008 (0.018)	0.023* (0.012)
Constant	0.262* (0.142)	0.371** (0.145)
Joint significance p-value	0.489	0.124
Observations	1783	1708
R ²	0.247	0.149

Note: Table shows the correlates of contract switches. Column (1) is estimated using the subset of treated units switching from scenario 1 to 3 and column (2) the subset switching from scenario 1 to 4. An observation is a contracting unit-half year, where a contracting unit is defined as an insurer-practice-woman type. A woman type is a combination of age group (age ≤ 27 , age > 27) and pregnancy risk (high vs. low). Specifications include insurer-woman type, insurer-practice, and half year fixed effects. Standard errors in parentheses are clustered at the contracting unit level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$.

5 Results

Figure 4 presents our main event study results for the impact of FFS contracts on c-section rates and Appendix Figure 2 presents raw outcome trends. Panel A examines switches from BP to FFS for c-sections, holding vaginal deliveries under BP. We find that this switch increases c-section rates 17 p.p. (34%) relative to baseline, with effects persisting for up to two years after the switch. The dynamic treatment effects are relatively flat in the post-period, suggesting staggered contract switches are likely unrelated to underlying woman risk or practice characteristics. Panel B evaluates the effect of switching vaginal delivery contracts from BP to FFS while keeping c-section contracts at BP. We find that the c-section rate falls 11 p.p. (22%) on average and that treatment effects are increasing over time. In both panels, we see parallel trends prior to the switch.

FIGURE 4: Impact of Contracts on C-section Rates



Note: Figure shows coefficients and 95% confidence intervals of event studies using c-section rates as the outcome variable. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1 where both procedures are covered under BP. The text inside each panel reports the average post-switch coefficient and its standard error in parenthesis. Standard errors are clustered at the insurer-practice-woman type level.

The aggregate trends presented in Figure 3 showed that the use of FFS falls for both c-sections and vaginal deliveries during the sample period and that, correspondingly, c-section rates decline. Based on our estimates, a simple back-of-the-envelope calculation would suggest that 6 p.p. of the decline in c-section rates may be explained by the move

from FFS to BP for c-sections (= -17 p.p. for c-section switches + 11 p.p. for vaginal delivery switches).¹³

5.1 Marginal Women

Our stylized model in Section 2 predicts that contract changes that increase c-section rates do so by encouraging providers to perform c-sections on marginally sicker women. Sicker women are more likely to be unprofitable under a BP contract relative to their healthier counterparts because they typically require more ancillary services which have zero marginal prices. Hence, the switch to FFS is expected to increase utilization by a greater magnitude among women in worse health.

To test these predictions, we explore the heterogeneity in treatment effects by several measures of patient health calculated using pre-period averages at the insurer-practice level. The key advantage of this approach is that pre-treatment health measures are immune to any endogenous reclassification of women into high-risk status that may occur in response to the contract switch. We use the fraction of women with high-risk pregnancies, with above-average age, with chronic diseases, and with placenta previa/abruptia.¹⁴

Figure 5, Panel A reports event study estimates stratified by whether the fraction of high-risk pregnancies is above (dark blue) or below (light blue) the median of its distribution (weighted by the pre-period number of deliveries). Consistent with the model's predictions, Panel A1 shows that insurer-practice pairs with relatively low prevalence of high-risk pregnancies experience an increase in c-section rates of approximately 8 p.p. following the switch of c-sections from BP to FFS, whereas pairs with a high prevalence exhibit a substantially larger increase, of nearly 25 p.p. In this latter group, however, we see evidence of significant pre-trends ($p < 0.01$), so these estimates should be interpreted

¹³This back-of-the-envelope calculation holds under the assumptions that (i) switching c-section contracts from FFS to BP has the same impact in magnitude but of different sign than the switch from BP to FFS, (ii) the only contract switches observed in the country are from scenarios 1 to 3 and from scenarios 1 to 4, and (iii) there is an equal number of deliveries represented by the switch from scenarios 1 to 3 as there are by switch from scenarios 1 to 4.

¹⁴We implement all heterogeneity analyses using the estimator of [Borusyak et al. \(2024\)](#) with the `hetby` option, interacting the period fixed effects with the grouping variable so that counterfactuals for each group are imputed only from contracting units in that group. Because the command reports a single set of pre-period coefficients pooled across groups, we recover group-specific pre-trends by estimating separate regressions on the pre-period observations of each group.

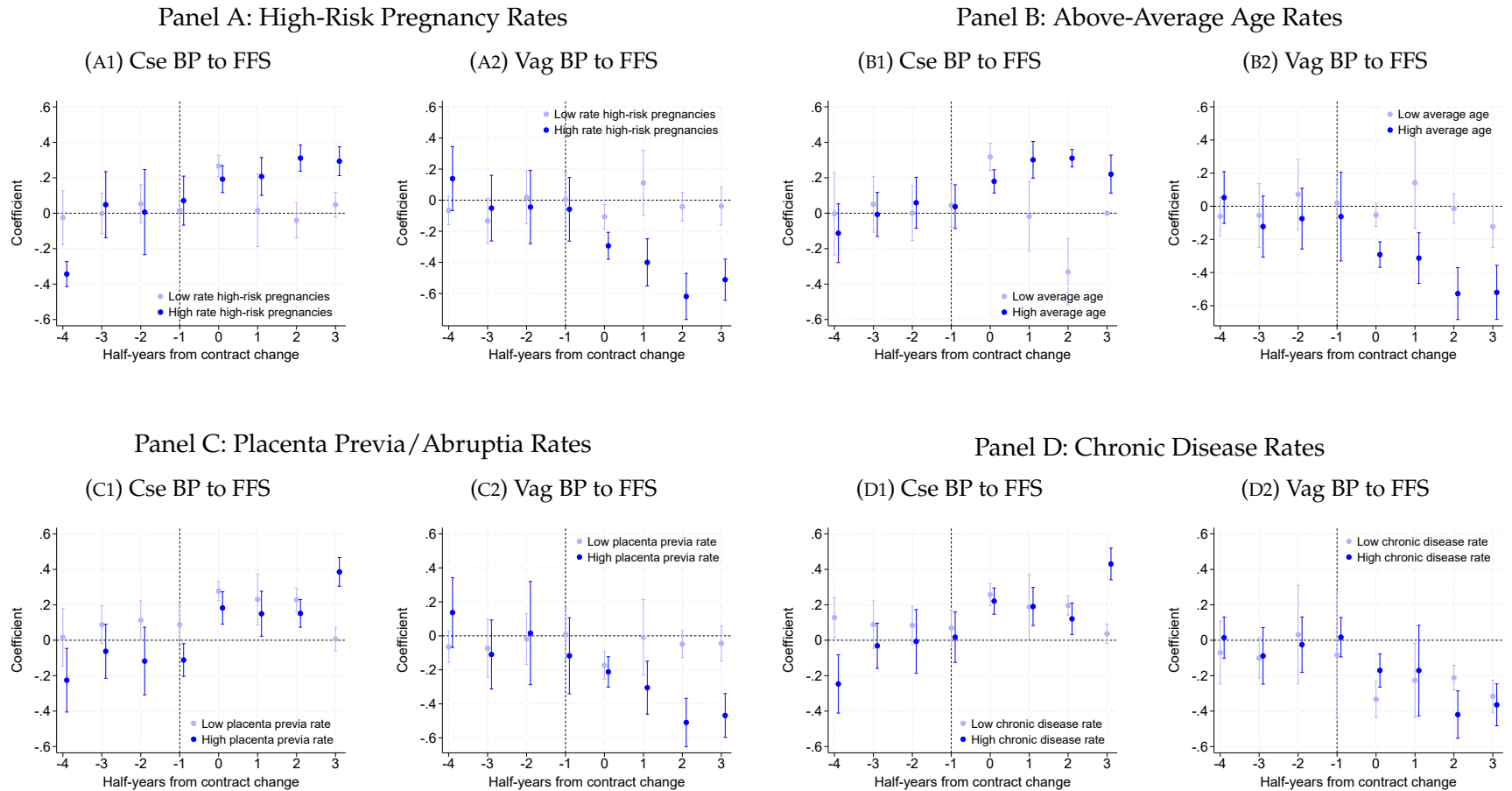
with caution. Reassuringly though, results in Panel A2 where we see strong evidence of parallel pre-trends are qualitatively similar: the reduction in c-section rates after the switch of vaginal deliveries from BP to FFS is entirely explained by insurer-practice pairs with a high prevalence of high-risk pregnancies.

Panels B and C present results by the pre-period fraction of women with above-average age and by the prevalence of placenta previa/abruptia, respectively. In each panel, dark blue estimates correspond to insurer-practice pairs with risk measures above the median (weighted by the pre-period number of deliveries), and light blue estimates to pairs below it. We again find that treatment effects are nearly twice as large among insurer-practice pairs serving higher-risk women.

The placenta previa/abruptia results are particularly informative. Panel C2 shows substantially larger reductions in c-section rates following the switch of vaginal deliveries to FFS among pairs with high pre-period rates of placenta previa/abruptia, indicating that where clinical need for c-sections is more common, the procedures forgone are concentrated among women better suited for them. Panel C1 shows no significant differences between the two groups for the c-section contract switch, though the pre-trend test rejects for the high placenta previa/abruptia group ($p = 0.02$), so we treat this comparison as suggestive. Given that c-sections are indicated for women with serious pregnancy complications like placenta previa/abruptia, these results emphasize that financial incentives can be designed in a way that aligns with diagnostic skill and clinical suitability.

Finally, Panel D shows no statistically significant differences by chronic disease prevalence. One explanation is that the chronic conditions we measure are broad, difficult to target, and less directly tied to the clinical case for a c-section. The heterogeneity we detect appears specific to measures of obstetric risk.

FIGURE 5: Impact of Contracts on C-section Rates by Health Status

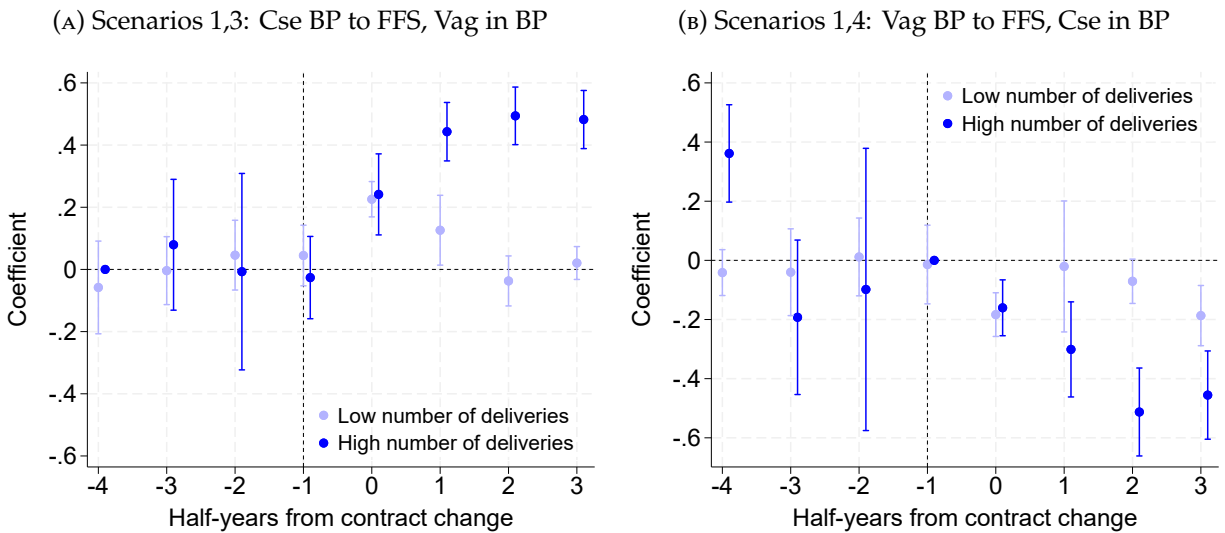


Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. Panel A presents heterogeneity results by pre-period high-risk pregnancy rates at the insurer-practice level. Panel B presents heterogeneity by pre-period prevalence of women over the age of 27 at the insurer-practice level. Panel C presents heterogeneity by rates of placenta previa/abruptia at the insurer-practice level. Panel D presents heterogeneity by pre-period chronic disease rates at the insurer-practice level. In Panels A1–D1, treated units switch c-sections from BP to FFS holding vaginal delivery fixed at BP (Scenarios 1,3); in Panels A2–D2, treated units switch vaginal delivery from BP to FFS holding c-section fixed at BP (Scenarios 1,4). Estimates in dark (light) blue use contracting units in the treatment groups above (below) the median of the relevant distribution weighted by the pre-period number of deliveries. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice level.

5.2 Marginal Practices

We now evaluate which healthcare practices are most responsive to the financial incentives created by FFS and BP contracts. Larger practices may be better able to respond to these incentives because they can invest in administrative capacity to monitor medical decisions. To see how our results interact with practice size, we estimate our event study specification separately for insurer-practice pairs with pre-period number of deliveries above and below the median.

FIGURE 6: Impact of Contracts on C-section Rates by Provider Size



Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. Estimates in dark (light) blue use contracting units in the treatment groups with pre-period number of deliveries above (below) the median percentile of the distribution weighted by the pre-period number of deliveries. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice level.

Figure 6 presents the results, with estimates for small insurer-practice pairs shown in light blue and for large pairs in dark blue. Across all contract switches, treatment effects on c-section rates are driven by large practices. For example, Panel A shows that a switch in c-section contracts from BP to FFS (holding vaginal delivery fixed at BP), increases the c-section rate among large practices by 32 p.p. on average, while among small practices the effect equals 13 p.p. Similarly, Panel B shows that switching vaginal delivery contracts from BP to FFS (while holding c-sections at BP) lowers c-section rates by 15 p.p. in large practices, compared to 7 p.p. in small ones. Although there is evidence of differential

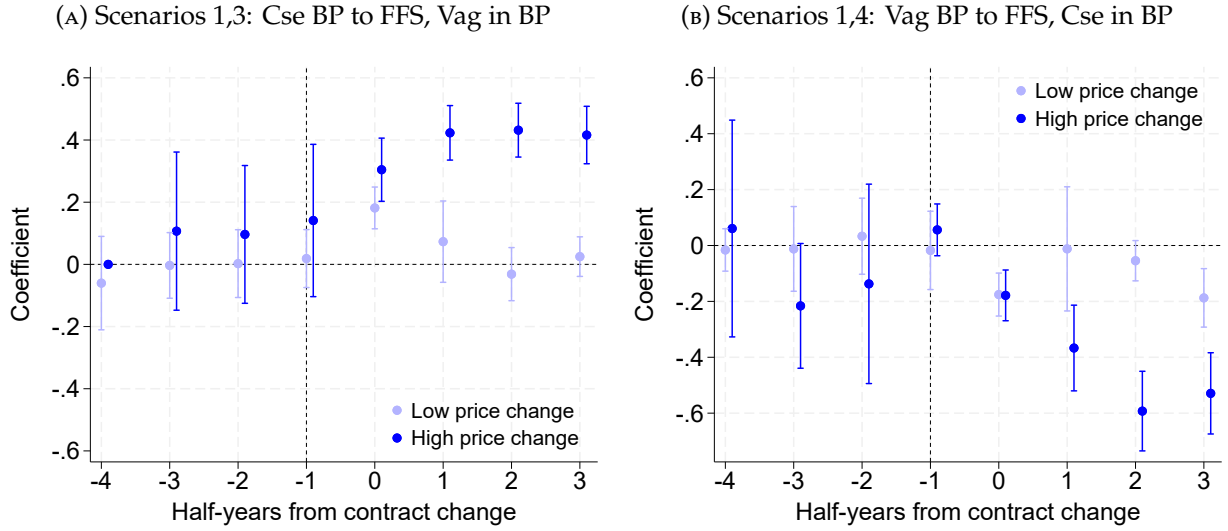
pre-trends among large practices in this exercise, Appendix Figure 4 corroborates our results: treatment effects weighted by the number of deliveries (giving more weight to larger practices) are similar to those in our main specification, suggesting larger practices drive the effects in both types of contract switch.

5.3 Price Levels versus Nonlinearities in Prices

The finding that the switch to FFS increases utilization of the procedure undergoing the contract change suggests that, on average, the potential profit gains from separately pricing ancillary services more than compensate the reduction in profits from potentially lower delivery procedure prices. As shown in Section 2, this is the condition required for the switch to FFS for c-sections to increase c-section rates.

Our goal in this section is to determine the extent to which each mechanism (delivery price levels under FFS and zero prices for marginal ancillary services under BP) affects medical decisions. For the first mechanism, we estimate the heterogeneity in treatment effects by predicted FFS prices. As seen in Figure 1, Panels A and B, our prediction is that the change in c-section rates after a switch to FFS is higher the higher is the FFS price. To implement this exercise, for each treated contracting unit, we calculate the expected price of the procedure undergoing the contract change to FFS. This prediction equals the average FFS price that the insurer pays for the procedure among contracting units that are always under scenario 2 (where both c-sections and vaginal deliveries are covered under FFS). For example, if insurer j and practice h switch from scenario 1 to 3, we assign the average c-section FFS price that j pays units in scenario 2. Then, we calculate the percentage change between the original BP price and the predicted FFS price, and estimate our event study specification separately for insurer-practice pairs with above- versus below-median percentage changes in price (weighted by the pre-period number of deliveries).

FIGURE 7: Impact of Contracts on C-section Rates by Predicted Change in Price Levels

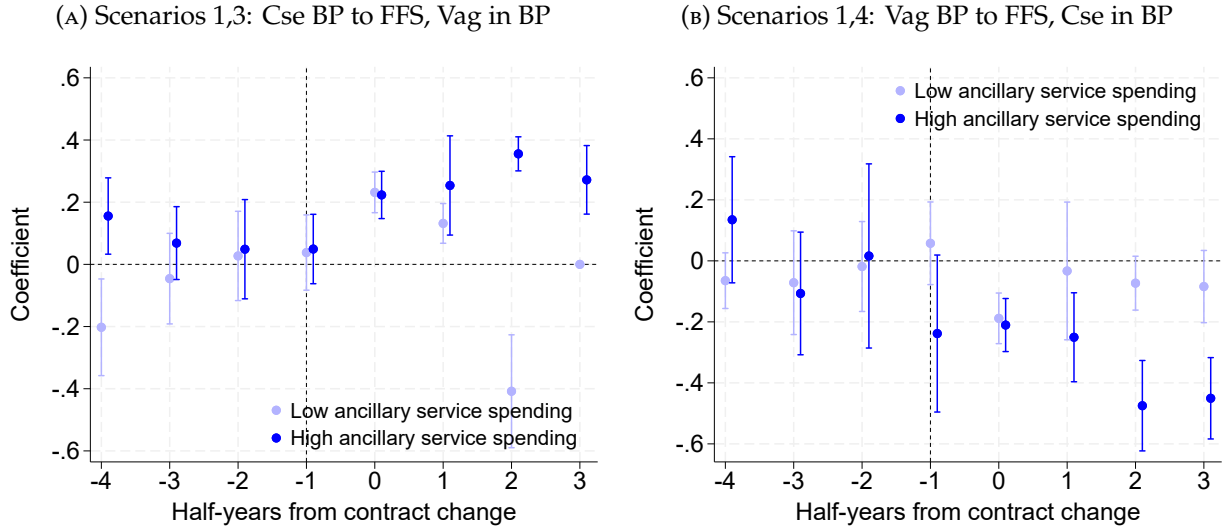


Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. Estimates in dark (light) blue use contracting units in the treatment groups for which we predict above-average (below-average) changes in procedure prices after the switch. The predicted price for a procedure after the switch equals the average price among contracting units always in scenario 2 and for the same insurer. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

Figure 7 presents the results. In line with our predictions, in Panel A, we estimate a relatively small increase in c-section rates (equal to 11 p.p. on average) among insurer-practice pairs with below-average predicted price changes for c-section (in light blue). In contrast, among insurer-practice pairs with above-average changes in c-section prices (in dark blue), the c-section rate increases by nearly 30 p.p. Panel B also shows a 14 p.p. reduction in c-section rates among insurer-practice pairs with above-average predicted price changes for vaginal delivery after the switch to FFS (in dark blue), but only a 7 p.p. decrease among their counterparts (in light blue).

For the second mechanism, we examine heterogeneity in treatment effects by pre-period spending on ancillary services. As illustrated in Figure 1, higher spending on these services magnify the response to the contract switch. We calculate the average pre-period healthcare spending on ancillary services under BP for each insurer-practice pair and compare event study results estimated separately on pairs with spending above or below the median of the distribution weighted by the pre-period number of deliveries.

FIGURE 8: Impact of Contracts on C-section Rates by Ancillary Service Spending



Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. Estimates in dark (light) blue use contracting units in the treatment groups for which pre-period spending on ancillary services is above (below) the 90th percentile of the distribution. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

Figure 8 reports the event study estimates. The results are consistent with the hypothesis that nonlinear pricing (i.e., zero prices for marginal ancillary services) induces meaningful distortions in medical treatments. In Panel A, we find that among insurer-practice pairs with relatively high spending on ancillary services, the average estimated effect of switching c-section contracts from BP to FFS equals 27 p.p., compared to 1 p.p. for those with relatively small spending. In Panel B, we also find significantly larger treatment effects for switches in vaginal delivery contracts among insurer-practice pairs with high baseline spending on ancillary services – those for which BP is particularly costly.

Together, Figures 7 and 8 indicate that both price levels and nonlinearities in pricing affect medical decisions in the context of childbirth. Distinguishing between these two margins matters for policy. If providers respond only to price levels, then policies that regulate prices directly, such as price caps or making delivery services free to women, could effectively manage incentives for overuse. However, if nonlinearities also matter, price regulation alone may be insufficient, and policies targeting the structure of contracts themselves may prove more effective. More broadly, our results imply that summarizing

contracts with a single linear price in this setting may miss important incentives created by the full structure of contract terms.

5.4 Robustness Checks

We conduct several robustness checks to assess the validity of our identification strategy and support the causal interpretation of our results. Appendix Figure 3 re-estimates our event study excluding the control group, so that identification relies solely on the staggered timing of contract switches among treated contracting units. Results for the switch of c-sections from BP to FFS are robust to this restriction. Estimates for the corresponding switch of vaginal deliveries display the same qualitative pattern, though treatment effects are significant only in the first year after the switch. Appendix Figure 4 presents event study estimates that weight each observation by its pre-period number of deliveries. The weighted specification yields similar effects as our main one, indicating that our results are not driven by outlier insurer-practice pairs.

To address potential concerns about endogenous changes in the composition of women, we implement two alternative sample restrictions: (i) women who remain with the same insurer throughout the sample period and (ii) women who choose their delivery practice in the first trimester of their pregnancy and made at least two claims there, helping ensure that delivery practice choice is not influenced by delivery contracts. In Appendix Figure 5, we see that both sample restrictions generate similar results as our main ones.

To further assess whether contract changes influence patient or practice composition, we estimate our event study using per capita maternal characteristics and practice characteristics as outcomes. Appendix Figures 6 and 7 show that contract switches are generally unrelated to these characteristics, reinforcing the conclusion that selection into contracts is minimal. The main exception is the switch of vaginal deliveries from BP to FFS, where a few practice characteristics exhibit modest pre-trends.

In Appendix Figure 8, we present event study results including insurer-practice and contracting unit fixed. These alternative specifications yield similar results as our main specification in Panel A, but more conservative estimates in Panel B. As additional robust-

ness checks we consider: excluding observations for which the modal predicted contract does not equal the observed contract in Appendix Figure 9, excluding municipalities where a large insurer that was terminated by the government in 2015 had more than 40% market share in Appendix Figure 10 (see [Serna \(2025\)](#)), using the ages of 20 and 30 as alternative cutoffs to determine contracts in Appendix Figure 11, and changing our time periods from half years to quarters in Appendix Figure 12.

5.5 Risk Selection

As discussed in Section 2, healthcare contracts may influence incentives to engage in both risk selection and moral hazard. Under moral hazard, conditional on the pool of enrollees, healthcare practices may choose to perform the most profitable procedure. Under risk selection, insurers with disproportionate use of FFS may choose to enroll women with lower risk scores or lower ex-ante healthcare costs compared to insurers with disproportionate use of BP. Our exercises so far show evidence of substantial moral hazard at the practice level, since c-section rates vary significantly with the contracts conditional on enrollment and women's characteristics.

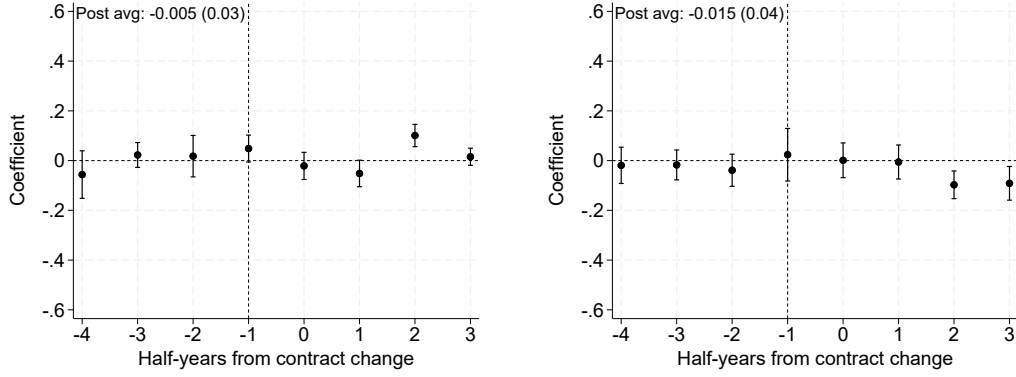
In this subsection, we explore whether contracts generate outcomes consistent with risk selection on the extensive and intensive margins. On the extensive margin, we assess whether contracts influence women's (discrete) decision to enroll with a particular insurer. On the intensive margin, we follow the approach of [Brown et al. \(2014\)](#) to evaluate whether contracts incentivize costlier women to enroll.

Figure 9 shows event study estimates using the average per capita likelihood of switching insurers and lagged per capita healthcare spending as outcomes. We find no systematic evidence that contract switches encourage women to switch insurers or encourage cheaper women to enroll. Panels A1 and A2 illustrate the effect of switches in c-section and vaginal delivery contracts from BP to FFS, respectively. We see very precise null effects on switching rates. In Panel B1, results also show no systematic patterns for lagged healthcare spending following the contract switch. However, Panel B2 shows an increase in lagged healthcare spending two years after the switch.

FIGURE 9: Impact of Contracts on Risk Selection Incentives

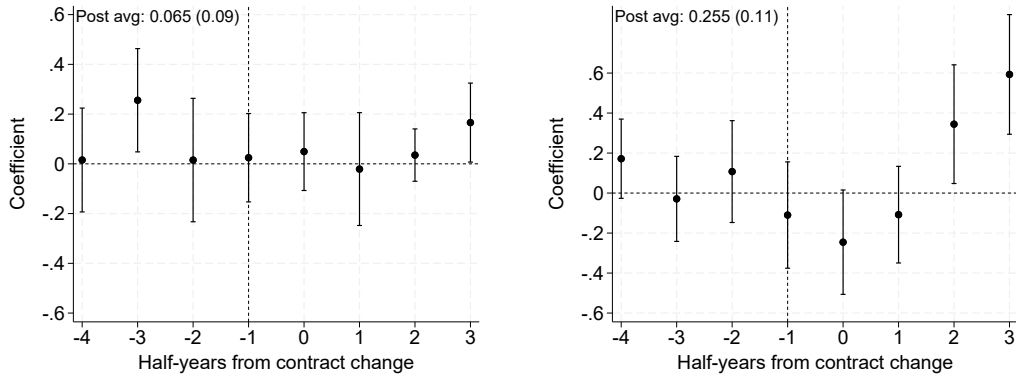
Panel A: Switching Rates

(A1) Scenarios 1,3: Cse BP to FFS, Vag in BP (A2) Scenarios 1,4: Vag BP to FFS, Cse in BP



Panel B: Lagged log healthcare spending

(B1) Scenarios 1,3: Cse BP to FFS, Vag in BP (B2) Scenarios 1,4: Vag BP to FFS, Cse in BP



Note: Figure shows coefficients and 95% confidence intervals of event studies using the insurer switching rate in Panel A and the log of lagged per capita healthcare spending in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

Generally we view these findings as being consistent with limited selection on both the extensive and intensive margins and support the plausibly exogenous nature of contract variation within insurer-practice-woman types. Moreover, the results indicate that the observed changes in procedure choice and care intensity are driven primarily by practice moral hazard, rather than by endogenous changes in patient selection.

5.6 Health Outcomes

Given our findings that contracts significantly influence procedure choice, we turn to assessing whether these changes have downstream effects on maternal and infant health. Understanding the health consequences is especially important because public policy has long expressed concern about the potential “overuse” of c-sections among women with low clinical need for the procedure (California Health Care Foundation, 2022). The literature offers mixed evidence on the relationship between c-sections and maternal health outcomes (Card et al., 2023; Fischer et al., 2024), raising the question of whether financial incentives embedded in contracts may not only alter medical treatment patterns but also impact health.

TABLE 6: Impact of Contracts on Maternal and Infant Health Outcomes

	Low APGAR		SMM	
	Cse BP to FFS (1)	Vag BP to FFS (2)	Cse BP to FFS (3)	Vag BP to FFS (4)
<i>Panel A: Full sample</i>				
Treated $\times$ Post	-0.00214 (0.00455)	0.00524 (0.00418)	-0.00531 (0.0183)	-0.00563 (0.0127)
Contracting units	2271	2161	2271	2161
<i>Panel B: High-risk pregnancy rates</i>				
Treated $\times$ Post	-0.00505 (0.0100)	0.00280 (0.0113)	-0.0657 (0.0404)	-0.107* (0.0440)
Contracting units	711	681	711	681
<i>Panel C: Above-average age rates</i>				
Treated $\times$ Post	0.000126 (0.00638)	0.00647 (0.00635)	-0.0380 (0.0228)	-0.0742* (0.0341)
Contracting units	1141	1043	1141	1043

Note: Table shows difference-in-differences estimates using as outcomes the fraction of births with 5-minute APGAR score below 8 and the rate of severe maternal morbidity. Columns (1) and (3) show the effect of switching the c-section contract from BP to FFS holding vaginal delivery fixed at BP. Columns (2) and (4) show the effect for the vaginal delivery contract switch from BP to FFS holding c-sections fixed at BP. The control group are contracting units always under scenario 1. Panel A uses the full sample, Panel B the subsample of insurer-practice pairs with above-median high-risk pregnancy rates in the pre-period, and Panel C the subsample of insurer-practice pairs with above-median average age rates in the pre-period. Specifications include insurer-woman type fixed effects and half year fixed effects. Standard errors in parentheses are clustered at the insurer-practice-woman type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$.

In Table 6, we estimate equation (1) using the rate of severe maternal morbidity (SMM)

and the rate of 5-minute APGAR scores below 8 as dependent variables.¹⁵ Panel A shows no statistically significant effects in the full sample for either outcome or contract switch. Because our findings above indicate that treatment effects on c-section rates are concentrated among insurer-practice pairs serving higher-risk women, Panels B and C examine health outcomes within these subgroups, restricting to pairs with above-median high-risk pregnancy rates and above-median rates of women aged 27 or older, respectively. Effects on low APGAR rates remain small and insignificant throughout. For SMM, point estimates in these subgroups are negative, and the effect of switching vaginal deliveries to FFS is marginally significant. Recall that this switch reduces c-section rates; the negative point estimates are therefore consistent with fewer c-sections modestly reducing maternal morbidity among higher-risk populations, though given the marginal significance and the number of subgroups examined, we interpret these estimates as suggestive. Overall, we find no evidence that contract-induced changes in delivery procedures harm short-term maternal or infant health, and if anything, reductions in c-section rates among high-risk groups may be accompanied by lower maternal morbidity.

6 Conclusions

A rich theoretical literature characterizes the incentives that arise in vertical relationships governed by linear prices, and in healthcare specifically, how insurer-provider payment arrangements shape treatment incentives. In practice, however, contracts between insurers and healthcare practices frequently feature nonlinear terms whose effects on downstream production decisions remain poorly understood. Empirical evidence is limited in part because studies lack detailed information on the contracting unit, the timing of contracts, and in many cases the contracts themselves. With our data and setting, we are able to overcome some of these limitations.

In this paper, we estimate the causal effect of fee-for-service (FFS) and bundled payment (BP) contracts on the decision to provide c-sections versus vaginal deliveries. These

¹⁵Infants with low 5-minute APGAR scores are substantially more likely to develop cerebral palsy and other developmental disorders, making this score a standard measure of infant health (e.g., [Razaz et al., 2019](#); [Thavarajah et al., 2018](#)).

contract types differ along two margins: the level of the negotiated procedure price and the marginal price of ancillary services, which is positive under FFS and zero under BP. Because insurers and practices in Colombia may cover the two delivery procedures under different contracts, the downstream firm's incentive to supply each procedure depends on the full menu of contract terms it faces and not just on any single price.

Using an event study design that leverages contract switches within insurer, practice, and woman type, we find that switching a delivery procedure from BP to FFS substantially increases its utilization. For example, switching c-sections contracts increases the c-section rate by 17 p.p., while switching vaginal delivery contracts reduces the c-section rate by 11 p.p. We show that these effects are explained by both price levels and nonlinear contract terms. However, despite the large shifts in medical decisions, we find no meaningful impacts on maternal and infant health.

Our results indicate that bilateral contracting decisions in healthcare causally affect the type, cost, and intensity of care that women receive. The distinction between price levels and nonlinear terms matters for how this market is regulated. Policies that target price levels address only one of the margins we identify. Effective regulation of medical practice through contracting may therefore require attention to both the level and the structure of negotiated payments.

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Appendix A Data Cleaning

The raw data contains all the delivery procedures for women enrolled with insurers participating in the contributory health care system during 2013 to 2019. The total number of observations in this data set is $N = 1,017,830$. To obtain our final analysis sample we proceed in the following steps:

1. Keep singleton, non-breech, first-time births ($N=851,464$).
2. Keep women in reproductive age ($N=849,775$).
3. Keep deliveries that are reimbursed under fee-for-service, capitation, or bundled payment ($N=800,807$).
4. Obtain the modal contract for each service (c-section and vaginal delivery), insurer, practice, type of woman, and half year. A type of woman is defined by a combination of age group (≤ 27 , > 27) and pregnancy risk. The age group cutoff represents the average age in the data.
5. Collapse the data to the insurer, practice, type of woman (contracting unit), and half year, retaining the modal contract and computing c-section rates and average per-capita measures of healthcare spending and of maternal and infant health outcomes.
6. For every contracting unit identify the first contract switch. Keep observations between the baseline contract and the next contract, and drop observations corresponding to subsequent contract switches. For example, if a contracting unit that has the following trend in contracting scenarios: (1, 1, 1, 3, 3, 2, 2), we keep only observations between the first and the fifth half-year defining the first switch from scenario 1 to 3.
7. Drop deliveries in 2013-1 for which contract switches are undefined ($N=778,510$).

APPENDIX FIGURE 1: Examples of Contracts Between Insurers and Providers

CONTRATO DE PRESTACIÓN DE SERVICIOS DE SALUD NÚMERO 148-2015

INFORMACION DE LAS PARTES QUE INTERVIENEN				
CONTRATANTE	ALIANZA MEDELLIN ANTIOQUIA EPS S.A.S		CONTRATISTA	E.S.E. HOSPITAL LA MISERICORDIA (VALI)
NATURALEZA JURIDICA:	MIXTA		NATURALEZA JURIDICA:	PÚBLICA

VALOR MONETARIO (V/M) COMPONENTE FIJO	EVENTO	TARIFA	PAQUETE INTEGRAL	TARIFA
	CONSULTA MEDICA GENERAL	SEGUN DECRETO 2423 VIGENTE MENOS 50%	ACTIVIDADES DE PROMOCION Y PREVENCIÓN, SEGUN LA NORMA TECNICA Y GUÍAS INTEGRALES DE LA RESOLUCION 432 DE 2006 DEL COMPONENTE DE SALUD ORAL (SELANTES, DENTARTRAJE, APLICACION DE FLUOR Y CONTROL DE PLACA)	PROPIAS (SEGUN ANEXO)
	HOSPITALIZACION DEL PRIMER NIVEL DE ATENCION	SEGUN DECRETO 2423 VIGENTE MENOS 50%		
	ATENCION INICIAL DE URGENCIAS, ESTABILIZACION, RESOLUCION O REMISION DEL PACIENTE EN URGENCIAS	SEGUN DECRETO 2423 VIGENTE MENOS 50%		
	ATENCION DEL PARTO DE BAJA COMPLEJIDAD	SEGUN DECRETO 2423 VIGENTE MENOS 50%		

INFORMACION DE LAS PARTES QUE INTERVIENEN					
CONTRATANTE	ALIANZA MEDELLIN ANTIOQUIA EPS S.A.S.	CONTRATISTA	ESE HOSPITAL SAN CAMILO LELIS - VEGACHI		
NATURALEZA JURIDICA:	MIXTA	NATURALEZA JURIDICA :	PUBLICA		
NIT:	900604350-0	NIT:	800114286-1		
SERVICIOS, ACTIVIDADES Y TECNOLOGIA EN SALUD	INCLUYE		FORMA DE PAGO	TARIFA	OPORTUNIDAD
ATENCION DEL PARTO DE BAJO NIVEL DE COMPLEJIDAD	DERECHO DE SALA DE PARTO, HONORARIOS MEDICOS Y TSH NEONATAL CUANTITATIVO		PAQUETE INTEGRAL DE ATENCION	\$698.800	APLICA TRIAGE II 30 MINUTOS, 1 DIA

Servicio	1. Baja	2. Media	3. Alta	No Aplica	Total
MEDICINA GENERAL	10.578	368	722	103	11.595
LABORATORIO CLÍNICO	7.013	2.561	1.352	230	10.348
ENFERMERÍA	9.089	334	636	254	10.045
ODONTOLOGÍA GENERAL	8.477	227	485	111	9.139
TOMA DE MUESTRAS DE LABORATORIO CLÍNICO	7.965	289	593	325	9.015
PSICOLOGÍA	7.093	408	587	42	8.001
SERVICIO FARMACÉUTICO	4.824	1.901	1.373	15	7.526
FISIOTERAPIA	5.185	257	497	321	6.198
NUTRICIÓN Y DIETÉTICA	4.966	313	458	48	5.735
GINECOBSTERICIA	509	4.339	357	135	5.111
SERVICIO DE URGENCIAS	3.811	878	479		4.943
TERAPIA RESPIRATORIA	4.053	180	396	218	4.806
MEDICINA INTERNA	222	3.994	349	30	4.537
PEDIATRÍA	384	3.666	317	45	4.257
HOSPITALIZACIÓN ADULTOS	1.873	825	813	646	3.979
TRANSPORTE ASISTENCIAL BÁSICO	3.803	83	98		3.976
FONOAUDILOGÍA Y/O TERAPIA DEL LENGUAJE	3.063	180	323	248	3.759
VACUNACIÓN	2.752	21	306	709	3.746
TOMA DE MUESTRAS DE CUELLO UTERINO Y GINECOLÓGICA	1.587	279	322	1.578	3.714
OBSTETRICIA	2.774	665	380		3.669
CIRUGÍA GENERAL	383	3.186	381	44	3.651
ORTOPEDIA Y/O TRAUMATOLOGÍA	166	3.211	293	24	3.645
ANESTESIA	107	3.123	300	37	3.544
PROTECCIÓN ESPECÍFICA - VACUNACIÓN	3.376	65	99	1	3.535
IMÁGENES DIAGNOSTICAS - IONIZANTES	1.110	1.342	717	620	3.532
PROTECCIÓN ESPECÍFICA - ATENCIÓN EN PLANIFICACIÓN FAMILIAR HOMBRES Y MUJERES	3.313	44	69		3.419
GENERAL ADULTOS	2.326	644	573		3.391
DETECCIÓN TEMPRANA - ALTERACIONES DEL EMBARAZO	3.109	47	72		3.226

Note: Figure shows screenshots of contracts signed between insurers and public hospitals in Colombia. The information was obtained from <https://consultorsalud.com/que-sabemos-contratos-act-de-eps-con-las-ips/>, https://www.hospitalyali.gov.co/wp-content/uploads/2015/05/doc_Contrato-Savia-Salud.pdf and <https://hfps.gov.co/plan-de-mercadeo-hfps-2024/>.

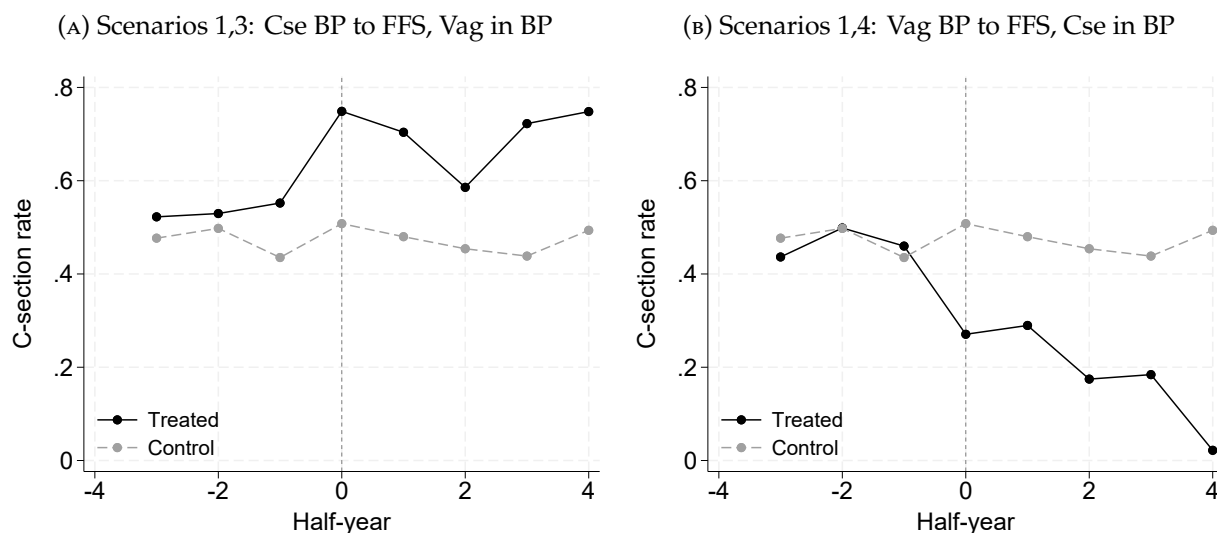
Appendix B Robustness Checks and Additional Results

APPENDIX TABLE 1: Correlation between Delivery Spending, C-sections, and Women's Health

	Log cost on delivery day
Unhealthy	0.005 (0.014)
C-section	0.421*** (0.011)
Unhealthy $\times$ C-section	0.071*** (0.021)
Constant	5.923*** (0.006)
Insurer-provider FE	Yes
Half year FE	Yes
Observations	33,577
R ²	0.524

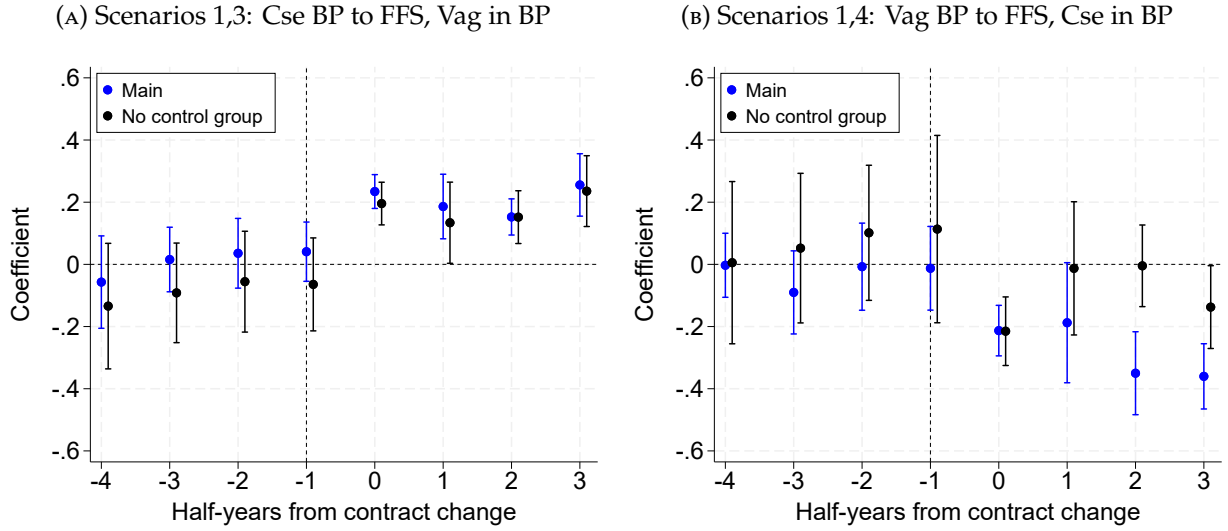
Note: Table shows OLS regression of the log of the cost on the delivery day (+ 1) on the interaction between an indicator for whether the woman has a chronic diseases and an indicator for whether she receives a c-section. An observation in this regression is a woman. We include insurer-provider and half year fixed effects. Standard errors are clustered at the woman level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.001$.

APPENDIX FIGURE 2: Raw C-section Rate Trends



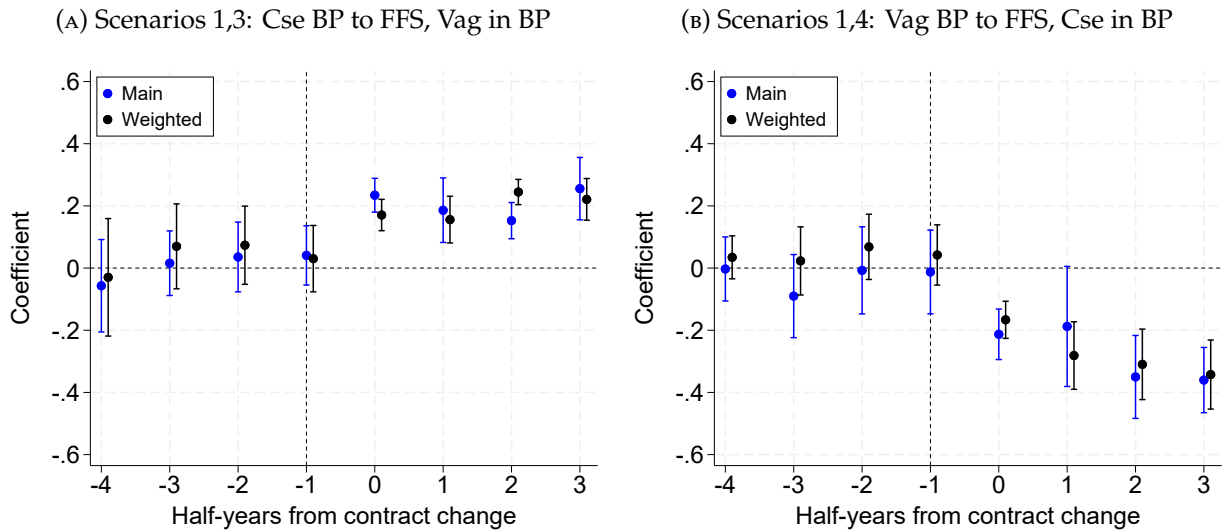
Note: Figure presents the average c-section rate among treated (in black) and control (in gray) groups. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In both cases, the control group are insurer-provider-woman types always under scenario 1. To construct this figure we assign a placebo switch date to the control group equal to the median switch rate in the treated group.

APPENDIX FIGURE 3: Impact of Contracts on C-section Rates Among Treated Units Only



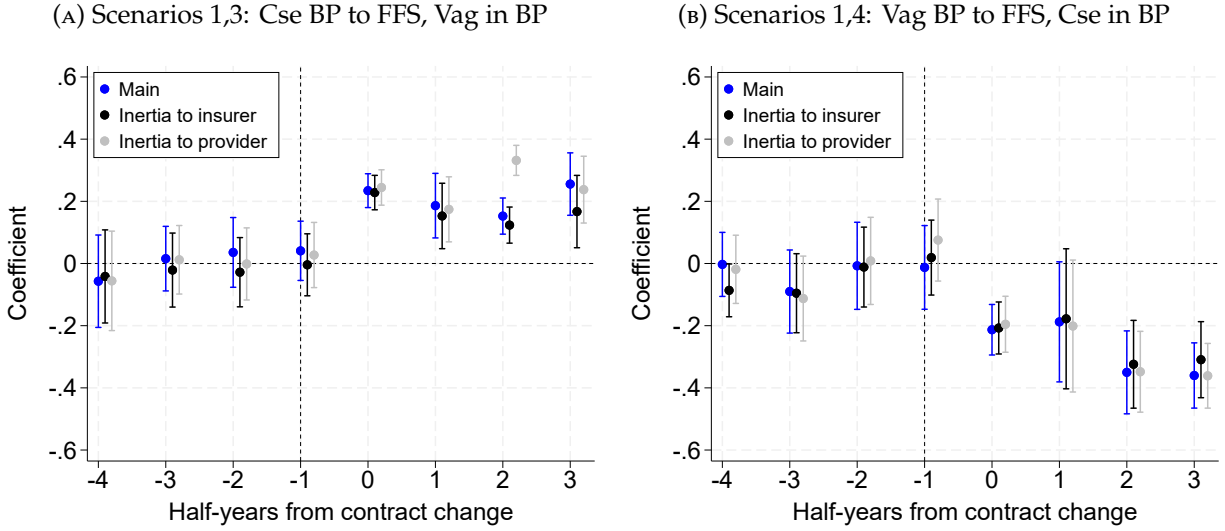
Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. We drop contracting units in the control group that do not experience contract switches. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. Standard errors are clustered at the insurer-practice-woman type level.

APPENDIX FIGURE 4: Impact of Contracts on C-section Rates Weighting by Pre-period Number of Deliveries



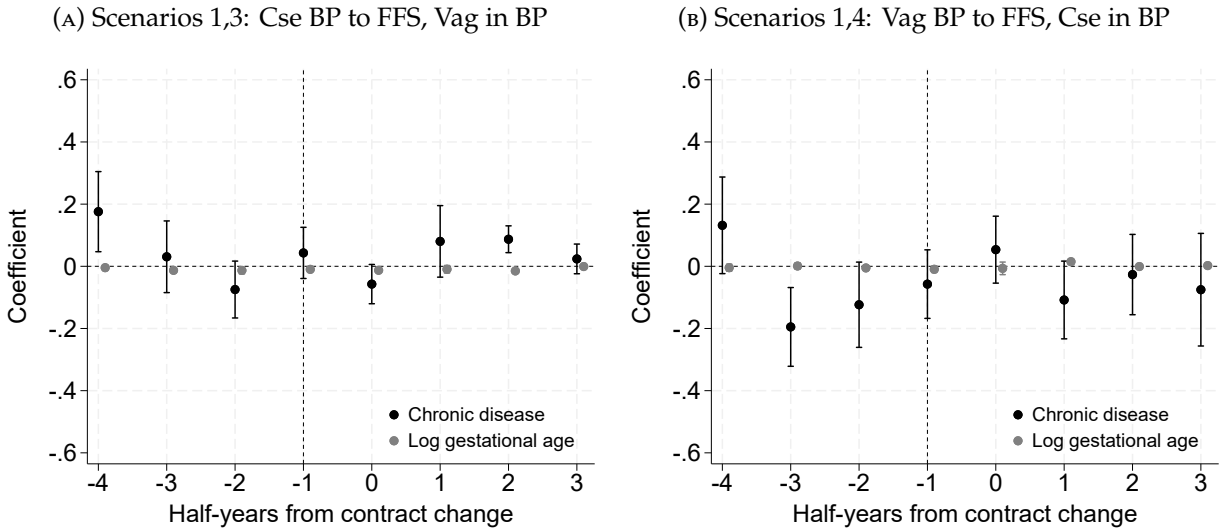
Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. We weight observations in the regressions by the average number of deliveries in relative time -1. Standard errors are clustered at the insurer-practice-woman type level.

APPENDIX FIGURE 5: Impact of Contracts on C-section Rates Among Women Inertial to their Insurer and their Provider



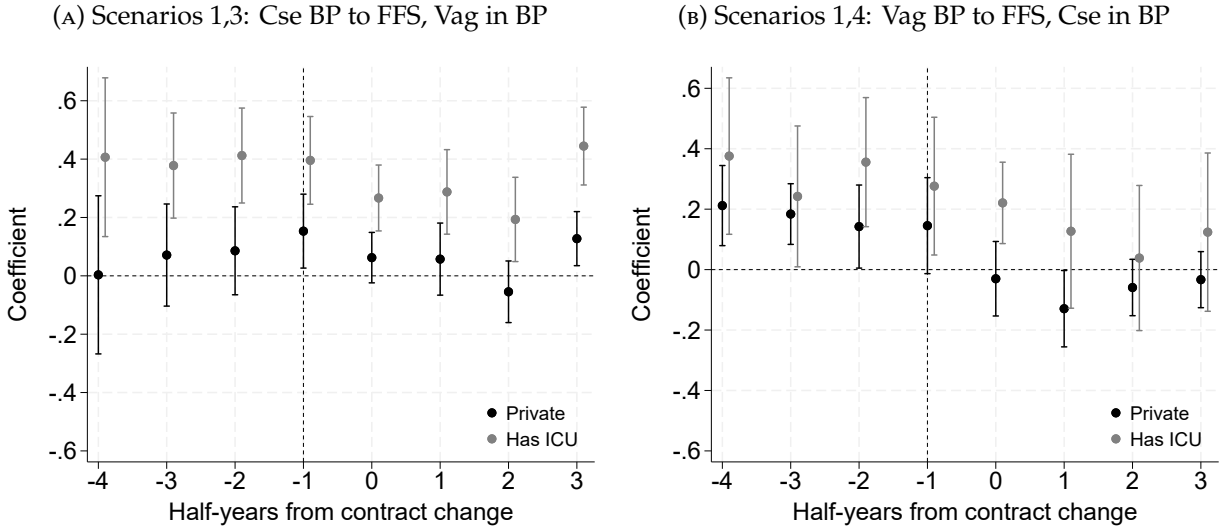
Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. We drop women who switch their insurer over the sample period. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

APPENDIX FIGURE 6: Impact of Contracts on Maternal Characteristics



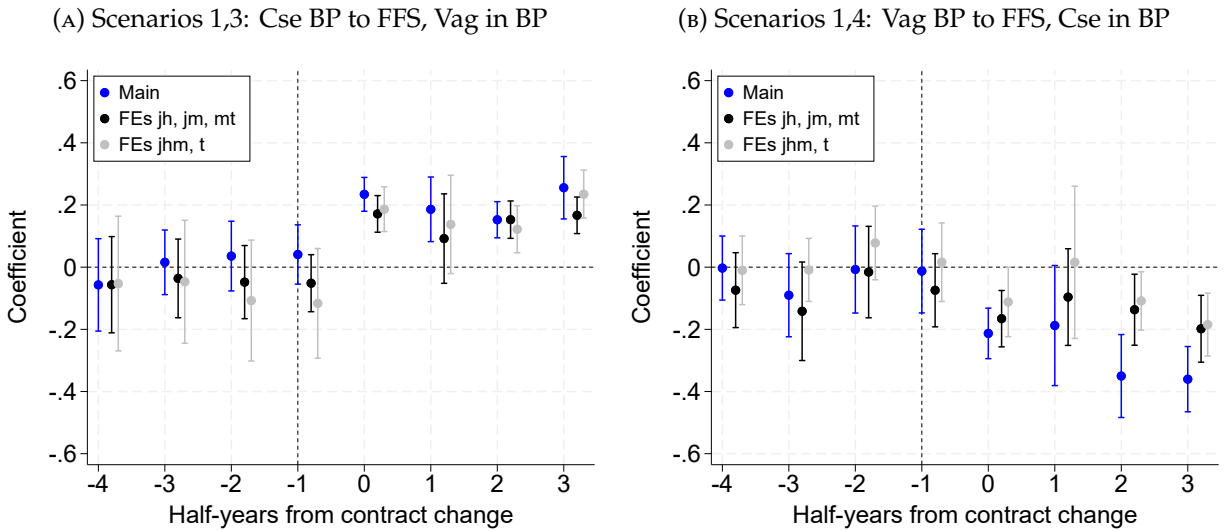
Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the chronic disease rate in black and the log of the average gestational age in dark gray. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

APPENDIX FIGURE 7: Impact of Contracts on Practice Characteristics



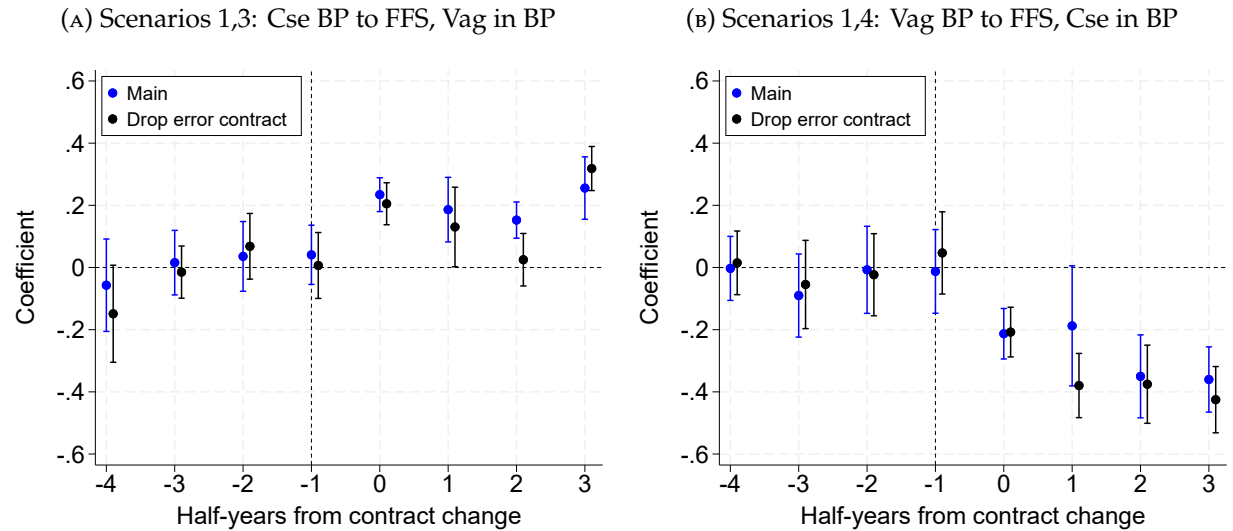
Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome an indicator for whether the practice is a private practice (in black) and whether the practice has either an intensive or intermediate care unit (in gray). The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

APPENDIX FIGURE 8: Impact of Contracts on C-section Rates with Different Fixed Effects

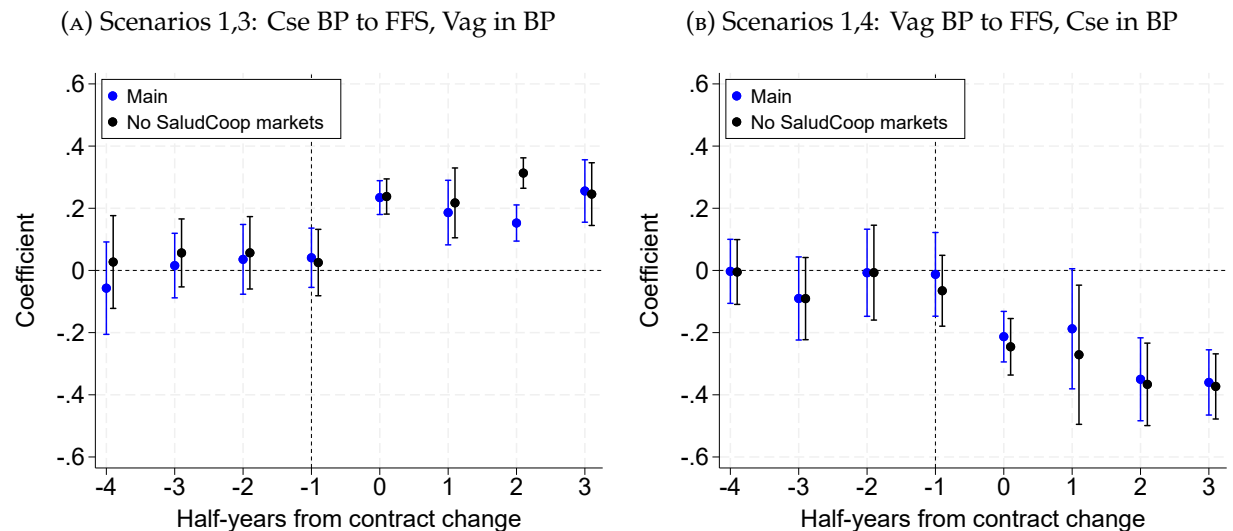


Note: Figure shows coefficients and 95% confidence intervals of event studies using as outcome the c-section rate. Results for specifications with insurer-provider, woman type, and half-year fixed effects are in black. Results for specifications with insurer-provider-woman type and half-year fixed effects are in gray. The treatment group is comprised of insurer-provider-woman types that: switch c-sections from BP to FFS while holding vaginal delivery fixed at BP in Panel A, and switch vaginal delivery from BP to FFS while holding c-section fixed at BP in Panel B. In all specifications, the control group are insurer-provider-woman types always under scenario 1. Standard errors are clustered at the insurer-practice-woman type level.

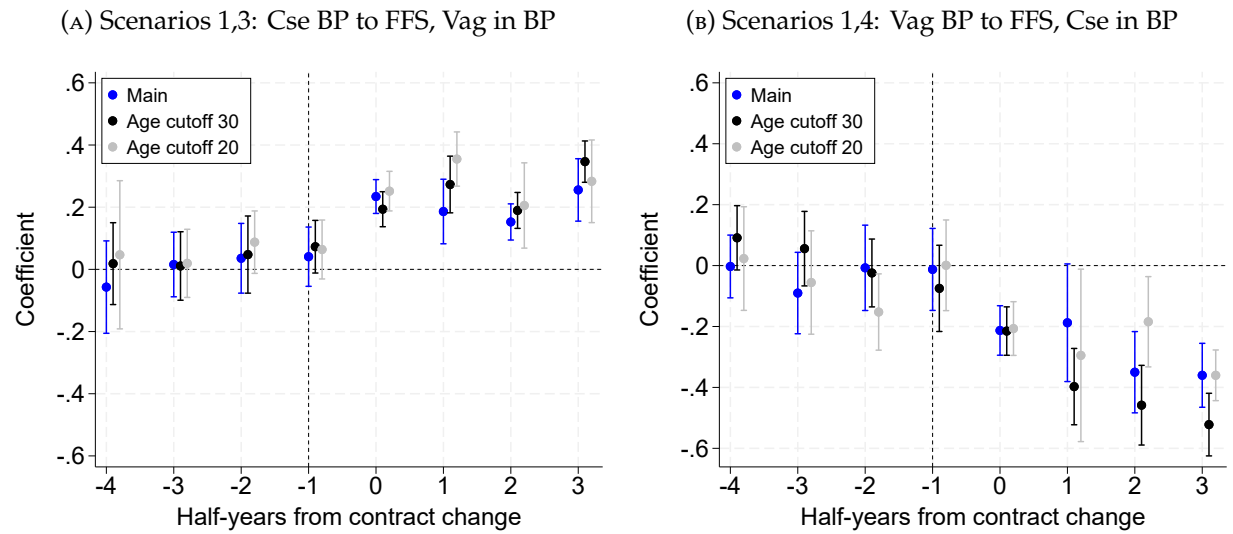
APPENDIX FIGURE 9: Impact of Contracts on C-section Rates Excluding Contracts that Do Not Equal Mode



APPENDIX FIGURE 10: Impact of Contracts on C-section Rates Excluding Markets with Presence of Terminated Insurer



APPENDIX FIGURE 11: Impact of Contracts on C-section Rates with Alternative Age Cutoffs



APPENDIX FIGURE 12: Impact of Contracts on C-section Rates with Alternative Time Periods

